

Singularities in Steady Water Waves using Analytic Continuation

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Abstract

This thesis studies the analytic structure of steady two-dimensional water waves using methods from complex analysis. For incompressible, inviscid, irrotational flows, the velocity field can be described by a complex potential that is analytic in the fluid domain. This analytic structure allows the use of conformal mapping and analytic continuation techniques to study the geometry of the free surface and the behaviour of the flow beyond the physical fluid region. A central result discussed in this work is Lewy's theorem [7] on analytic continuation across free boundaries, which explains why steady water wave profiles are locally analytic away from stagnation points. Building on this framework, the thesis examines the appearance of singularities in analytically continued solutions following the analysis of Tanveer [8]. In particular, singularities of square-root branch point type arise in the complex plane outside the physical domain and determine the radius of convergence of analytic representations of the free surface. The thesis also discusses the role of surface tension in the regularity and analytic continuation of water waves.

Popular scientific summary

Vattenvågor är ett naturfenomen som alla kan observera i en vattenmassa. Men deras matematiska beskrivning är mycket komplex. Inom matematiken behandlas vatten ofta som en inkompressibel och viskositetsfri vätska vars rörelse styrs av en uppsättning ekvationer som kallas Eulers ekvationer, medan formen på vattenytan måste bestämmas som en del av lösningen. I två dimensioner kan sådana flöden studeras med hjälp av verktyg från komplex analys. Denna uppsats studerar vattenvågor ur detta perspektiv och fokuserar på idén om analytisk fortsättning, där den matematiska beskrivningen av flödet kan utvidgas bortom vätskeregionen. Att studera denna utvidgning avslöjar förekomsten av singulariteter. Dessa är punkter som inte följer samma regler som de andra punkterna i området. Singulariteterna påverkar vågornas beteende vid vattenytan. Uppsatsen förklarar hur dessa singulariteter uppstår och hur de hjälper oss att bättre förstå strukturen hos vattenvågor.

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Chapter 1

Introduction

Water waves is a highly researched subject in fluid mechanics and mathematical analysis. They arise in many physical situations such as ocean surface waves, waves generated by wind, and disturbances created by moving objects in a fluid. From a mathematical perspective, water waves provide a rich example of a nonlinear free boundary problem in which the shape of the fluid domain is not known in advance but must instead be determined as part of the solution. Understanding the structure and regularity of such waves has therefore been one of the central areas of research in applied mathematics.

In the mathematical theory of water waves, the fluid is commonly modeled as inviscid, incompressible, and of constant density. Under these assumptions the motion of the fluid is governed by the Euler equations together with appropriate boundary conditions on the free surface separating the fluid from the surrounding air. The interface separating the fluid from the surrounding air is an unknown free boundary whose geometry must be determined as part of the solution. This coupling between the fluid motion and the shape of the boundary makes the problem nonlinear. A central goal in the theory is therefore to understand the regularity and geometric properties of the free surface as well as the behaviour of the fluid flow beneath it.

In two dimensions the water wave problem possesses an elegant mathematical structure when the flow is irrotational. In this case the velocity field can be written as the gradient of a scalar velocity potential satisfying Laplace's equation. The velocity potential and the associated stream function then form a pair of harmonic conjugates. Together they define the complex potential of the flow, which is analytic in the fluid domain. This complex-analytic formulation makes it possible to apply techniques from complex analysis and conformal mapping in the study of steady water waves.

One of the key advantages of this formulation is that it allows the fluid domain, which contains an unknown free boundary, to be represented through a conformal mapping from a simpler domain. Such transformations play an important role in the mathematical analysis of water waves and provide a powerful framework for studying the geometry of the free surface. In particular, analytic properties of the conformal mapping can be used to understand both the regularity of the free boundary and the behaviour of the flow in regions beyond the physical domain.

One of the main ideas in modern water wave theory is the use of analytic continuation. Although the flow is physically defined only within the fluid domain, the analytic functions describing the flow can often be extended into a larger region of the complex plane. Studying this analytic continuation provides insight into the mathematical structure of the solution and allows us to investigate the location and nature of singularities in the analytically continued functions. These singularities are quite important because they determine the convergence properties of analytic expansions describing the free surface and therefore influence the behaviour of the wave profile.

The analytic continuation of solutions across the free surface is closely related to the structure of the boundary conditions governing the fluid motion. In particular, Lewy [7] showed that when harmonic functions satisfy suitable analytic boundary relations, the functions extend analytically across the boundary. In the context of steady water waves this result provides a rigorous explanation for the analyticity of the free surface away from stagnation points. When the non-degeneracy condition required for this continuation fails, the analytic structure may break down, allowing singularities such as sharp crests to form.

Another important aspect of water wave theory concerns the role of physical effects such as gravity, surface tension, and vorticity. Gravity waves arise when gravity acts as the primary restoring force, while capillary waves are influenced by the effects of surface tension. The presence of surface tension modifies the dynamic boundary condition through the curvature of the free surface and can lead to stronger regularity properties of solutions. In particular, surface tension can lead to improved regularity of the free surface, and under suitable conditions the free surface is analytic. The presence of vorticity introduces additional complexity into the problem and leads to a broader class of wave solutions.

The purpose of this thesis is to study the analytic structure of steady water waves and to investigate the singularities that arise in their analytic continuation. The analysis focuses primarily on two-dimensional steady irrotational waves in deep water and follows a complex-analytic approach. Particular emphasis is placed on understanding how analytic continuation across the free surface arises from the boundary conditions and how singularities appear in the analytically continued solutions.

The analysis in this thesis is based on several important contributions in the mathematical theory of water waves. The analytic continuation across free boundaries is based on Lewy's theorem [7], which provides the foundation for the results in Chapter 3. The study of singularities in analytically continued solutions follows the work of Tanveer [8], which forms the basis of Chapter 4. The discussion of capillary effects and analyticity in Chapter 5 is informed by the results of Matei [10], together with related approaches developed by Crowdy. These works provide the theoretical framework underlying the developments presented throughout this thesis.

The thesis is organized as follows. Chapter 2 introduces the mathematical formulation of the water wave problem and reviews the necessary background from complex analysis and the implicit function theorem. Chapter 3 presents the principle of analytic continuation across free boundaries, focusing on Lewy's theorem and its application to steady irrotational water

waves. Chapter 4 studies the singularities that arise in analytically continued solutions, following the analysis of Tanveer, and shows that these singularities are typically branch points of square-root type in the complex plane. Chapter 5 discusses analyticity results for capillary waves and highlights the regularizing effect of surface tension on the free surface. Finally, Chapter 6 outlines possible directions for further research, including extensions of the analysis to water waves with vorticity.

Chapter 2

Mathematical Background

In this thesis, the flow is assumed to be two-dimensional and irrotational unless vorticity or rotational flow of the fluid is mentioned. Here, the fluid is inviscid, that is with zero viscosity, and incompressible, with constant density. Let $\mathbf{u} = (u, v)$ denote the velocity field of the fluid, where $u(x, y, t)$ and $v(x, y, t)$ are the horizontal and vertical velocity components, respectively. The incompressibility condition is then

$$\nabla \cdot \mathbf{u} = 0. \quad (2.0.1)$$

We consider a coordinate system (x, y) moving with the wave at the constant wave speed c . In this moving frame a travelling wave of permanent form appears steady, so all flow quantities are independent of time. A wave of permanent form appears steady. Hence, all flow quantities are time-independent and we have

$$\partial_t u = \partial_t v = 0. \quad (2.0.2)$$

We assume that the wave profile is periodic in x with a period, say L , and is symmetric with respect to a certain vertical line unless stated. For now, we take the axis of symmetry to be the y -axis and consider a flow of infinite depth (deep-water wave). The symmetry of the wave profile about the y -axis implies corresponding symmetry properties of the velocity field:

$$u(x, y) = u(-x, y), \quad v(x, y) = -v(-x, y). \quad (2.0.3)$$

Together with the spatial periodicity, this yields in particular that the vertical velocity vanishes on the vertical boundaries of one spatial period window,

$$v = 0 \quad \text{on } x = \pm L/2, \quad (2.0.4)$$

so that these boundaries behave like impermeable walls (no flow crosses them). In the fluid flow, the interface between the fluid and a vacuum is the free boundary determined by the fluid's behaviour. The free boundary is assumed to be a smooth curve, unless stated otherwise. By these assumptions, we are considering a flow in $\{(x, y); -L/2 < x < L/2, -\infty < y < \eta(x)\}$ where $\eta(x)$ is representing the free boundary.

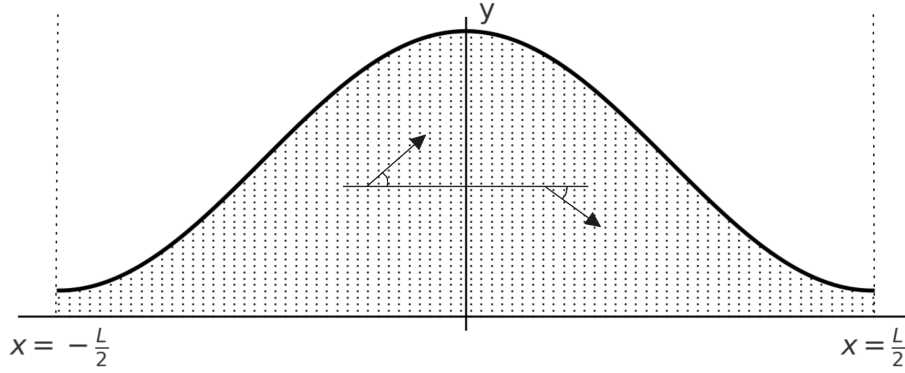


Figure 2.1: Symmetric wave [3]

Remark 2.0.1. In this formulation, for simplicity we will assume that the free surface can be represented as a graph of the form $y = \eta(x)$. This excludes more general wave profiles, such as pure capillary wave solutions (e.g. Crapper waves), where the free surface cannot be described by a single-valued function of x . The graph assumption is sufficient for the analysis carried out in this thesis.

The formulation of the governing equations and boundary conditions in this section followed standard treatments in the mathematical theory of water waves, and is based in part on the introduction given in Okamoto and Shoji [3].

2.1 Governing Equations for Inviscid Water Waves

We consider a two-dimensional travelling water wave of permanent form in infinite depth. The fluid occupies the domain

$$\Omega_\eta = \{z = x + iy \in \mathbb{C}; -L/2 < x < L/2, \quad -\infty < y < \eta(x)\}. \quad (2.1.1)$$

where $y = \eta(x)$ denotes the free surface. We assume that the wave profile is L -periodic and symmetric about the y -axis. Working in a frame moving with the wave speed c , the flow appears steady and all flow quantities are independent of time.

Since we are considering 2D flow, the incompressibility condition is

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \quad (2.1.2)$$

for the velocity vector $\mathbf{u} = (u, v)$. The Euler equations are,

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} \quad (2.1.3)$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial y} - g. \quad (2.1.4)$$

So, these time-dependent Euler equations are a system that models the flow of an inviscid, incompressible fluid with constant mass density ρ . The pressure of the fluid is p and g is the gravitational acceleration. Among these, only g and ρ are constants. So, in the two dimensional space, the PDE system for a wave can be written with the Euler equations, irrotationality condition (zero vorticity) and boundary conditions.

When the flow is rotational, we have vorticity defined as the curl of the fluid velocity field, represented by the equation $\vec{\omega} = \nabla \times \mathbf{u}$. It is a vector quantity that indicates the local rotation or spinning motion of a fluid element within the flow.

In the two-dimensional setting considered here, the scalar vorticity reduces to

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}. \quad (2.1.5)$$

If $\omega = \nabla \times \mathbf{u} = 0$, the flow is said to be irrotational. In this irrotational case, the vorticity of the flow vanishes.

When this condition holds in a simply connected domain, the velocity field can be expressed as the gradient of a scalar potential. Hence there exists a velocity potential function ϕ such that,

$$\mathbf{u} = \nabla \phi. \quad (2.1.6)$$

The velocity potential ϕ satisfies Laplace's equation: $\nabla^2 \phi = 0 = \Delta \phi$ for incompressible, irrotational, inviscid fluid flow with $\mathbf{u} = (\phi_x, \phi_y)$.

The boundary conditions on the free surface $y = \eta(x, t)$ are the kinematic and dynamic boundary conditions. The free surface consists of the same fluid particles over time. Therefore, the velocity of the fluid at the free surface must be tangent to the moving boundary. This gives the kinematic boundary condition,

$$\frac{\partial \phi}{\partial x} \frac{\partial \eta}{\partial x} = \frac{\partial \phi}{\partial y}, \quad (2.1.7)$$

The pressure on the free surface is prescribed by the surrounding air. Applying Bernoulli's law along the free surface then yields the dynamic boundary condition,

$$\frac{1}{2} |\nabla \phi|^2 + g\eta + \frac{\sigma}{\rho} \kappa = \text{constant}. \quad (2.1.8)$$

In the dynamic boundary condition, σ is the surface tension coefficient, ρ is the fluid density and κ is the curvature of the free surface.

If the free surface is given by $y = \eta(x, t)$, then the curvature κ of the surface is given by,

$$\kappa = -\frac{\eta_{xx}}{(1 + \eta_x^2)^{3/2}}. \quad (2.1.9)$$

The boundary condition on the bottom would depend on if we are working with finite or infinite depth,

at finite depth (say $y = -a$),

$$\frac{\partial \phi}{\partial y} = 0, \quad (2.1.10)$$

at infinite depth (as $y \rightarrow -\infty$),

$$|\nabla \phi| \rightarrow 0. \quad (2.1.11)$$

These conditions corresponds to the time-dependent formulation of the problem. In the case of steady travelling waves, we move to a frame travelling with the wave at constant speed c . In this frame the flow becomes steady and the condition is instead given by

$$\phi_x \rightarrow c, \quad (2.1.12)$$

$$\phi_y \rightarrow 0 \quad \text{as } y \rightarrow -\infty. \quad (2.1.13)$$

We will be working in infinite depth.

For two-dimensional incompressible flow it is also convenient to introduce a stream function $\psi(x, y)$ defined by

$$\mathbf{u} = (u, v) = (\psi_y, -\psi_x).$$

More mathematically, streamlines are curves in the fluid domain which are tangent at every point to the velocity field. Streamlines are the lines in which the fluid is flowing, forming the flow pattern.

Since

$$\nabla \psi \cdot \mathbf{u} = (\psi_x, \psi_y) \cdot (\psi_y, -\psi_x) = 0,$$

the stream function is constant along streamlines in the case of steady flow. Along a streamline,

$$\nabla \psi \cdot \mathbf{u} = 0. \quad (2.1.14)$$

In the steady travelling-wave setting, the free surface is transported with the flow and is therefore a streamline. Hence the stream function is constant on the free boundary $y = \eta(x)$:

$$\psi(x, \eta(x)) = \text{constant}. \quad (2.1.15)$$

This is a consequence of the kinematic boundary condition. Without loss of generality, we normalize the above constant to be zero and impose

$$\psi = 0 \quad \text{on } y = \eta(x). \quad (2.1.16)$$

Now, we know that the velocity potential ϕ and the stream function ψ can be related by the velocity field, that is

$$\mathbf{u} = (u, v) = (\phi_x, \phi_y) = (\psi_y, -\psi_x). \quad (2.1.17)$$

Thus,

$$\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y} \quad \text{and} \quad \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}. \quad (2.1.18)$$

These relations are the Cauchy–Riemann equations. Consequently, the complex function

$$W(z) = \phi + i\psi, \quad (2.1.19)$$

which is a function of a complex variable $z = x + iy$, is analytic in the fluid domain when the flow is irrotational and is called the complex potential. The velocity potential and stream function satisfy the Laplace’s equation.

For deep water, the flow approaches a uniform horizontal motion far below the surface. We know that $(u, v) \rightarrow (c, 0)$ as $y \rightarrow -\infty$, $u - iv$ is analytic and,

$$u - iv = \frac{dW}{dz} = \frac{\partial \phi}{\partial x} + i \frac{\partial \psi}{\partial x} = \frac{\partial \psi}{\partial y} - i \frac{\partial \phi}{\partial y}. \quad (2.1.20)$$

So in terms of the complex potential $W = \phi + i\psi$, $(u, v) \rightarrow (c, 0)$ as $y \rightarrow -\infty$ may be equivalently stated as,

$$\frac{dW}{dz} \rightarrow c \quad \text{as } y \rightarrow -\infty. \quad (2.1.21)$$

2.2 Complex analysis

In this section we take note of some complex-analytic notions from [2] that are essential for the analysis of steady two-dimensional water waves and which will be used throughout the thesis.

Definition 2.2.1 (Analytic function). Let $\Omega \subset \mathbb{C}$ be an open set. A function $f : \Omega \rightarrow \mathbb{C}$ is said to be *analytic* in Ω if it is complex differentiable at every point of Ω .

Definition 2.2.2 (Real analytic function). Let $\Omega \subset \mathbb{R}^n$ be an open set. A function $f : \Omega \rightarrow \mathbb{R}$ is said to be *real analytic* in Ω if, for every $x_0 \in \Omega$, there exists a neighbourhood U of x_0 and coefficients $a_\alpha \in \mathbb{R}$ such that

$$f(x) = \sum_{\alpha \in \mathbb{N}_0^n} a_\alpha (x - x_0)^\alpha, \quad x \in U.$$

Here

$$\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$$

is a multi-index, and

$$(x - x_0)^\alpha = (x_1 - x_{0,1})^{\alpha_1} \cdots (x_n - x_{0,n})^{\alpha_n}.$$

In two-dimensional potential flow, if $f = U + iV$ is analytic in Ω , then both U and V are harmonic in Ω , that is,

$$\Delta U = \Delta V = 0. \quad (2.2.1)$$

U, V are real analytic as functions of (x, y) if $z = x + iy$.

Definition 2.2.3 (Conformal map). Let $\Omega_1 \subset \mathbb{C}$. A function $z : \Omega_1 \rightarrow \mathbb{C}$ is called a *conformal map* if it is analytic and injective in Ω_1 , with $z'(\zeta) \neq 0$ for all $\zeta \in \Omega_1$.

Theorem 2.2.4 (Riemann mapping theorem). *Any non-empty simply connected open set $\Omega \subset \mathbb{C}$, distinct from $\mathbb{C}$ itself, is conformally equivalent to the open unit disk. That is, there is a conformal bijective map $z : \mathbb{D} \rightarrow \Omega$ where $\mathbb{D}$ is the open unit disk.*

This theorem justifies the use of conformal mappings to reformulate the free-boundary water-wave problem in a fixed domain. For periodic deep-water waves, one typically uses a conformal map

$$z = z(\zeta), \quad (2.2.2)$$

from a strip or the unit disk in the ζ -plane onto one period of the physical fluid domain, with the free surface corresponding to $|\zeta| = 1$. More details about this is further explained in Chapter 4.

Definition 2.2.5 (Analytic continuation). Let f be analytic in a domain $\Omega_1 \subset \mathbb{C}$. An *analytic continuation* of f to a larger domain $\Omega_2 \supset \Omega_1$ is an analytic function $\tilde{f} : \Omega_2 \rightarrow \mathbb{C}$ such that $\tilde{f}|_{\Omega_1} = f$.

Analytic continuation is a useful tool in water-wave theory. Although the flow is initially defined only in the physical fluid domain, the analytic functions describing the flow may admit continuations to larger regions of the complex plane. Studying these continuations can reveal singularities whose location and type contain information about the free surface and the convergence of series representations.

Theorem 2.2.6 (Principle of permanence of functional relations, [4]). *Let $\Omega \subset \mathbb{C}$ be a domain, let $z_0 \in \Omega$, and let*

$$\mathcal{F} : \Omega \times \mathbb{C}^{n+1} \rightarrow \mathbb{C}$$

be analytic. Suppose that f_0 is analytic in a neighbourhood $U_0 \subset \Omega$ of z_0 and satisfies

$$\mathcal{F}(z, f_0(z), f'_0(z), \dots, f_0^{(n)}(z)) = 0, \quad z \in U_0. \quad (2.2.3)$$

Let $\gamma : [0, 1] \rightarrow \Omega$ be a path with $\gamma(0) = z_0$ along which f_0 admits analytic continuation. Then every analytic element obtained by continuing f_0 along γ also satisfies the same relation. That is, if f denotes a continued branch near a point of the path, then

$$\mathcal{F}(z, f(z), f'(z), \dots, f^{(n)}(z)) = 0$$

wherever this continued branch is defined.

Note that this result is a simplified version of the principle of permanence of functional relations; see for instance Hille [4], Theorem 1.11.2.

Proof. Choose a chain of disks $D_0, D_1, \dots, D_m \subset \Omega$ covering the path γ , with $D_0 \subset U_0$, such that consecutive disks overlap. Let f_j denote the analytic element in D_j obtained by analytic continuation.

For each j , define

$$G_j(z) = \mathcal{F}(z, f_j(z), f'_j(z), \dots, f_j^{(n)}(z)).$$

Since $\mathcal{F}$ and f_j are analytic, each G_j is analytic in D_j . By assumption, $G_0 \equiv 0$ in D_0 .

Suppose now that $G_{j-1} \equiv 0$ in D_{j-1} . On the overlap $D_{j-1} \cap D_j$, the analytic elements f_{j-1} and f_j agree by the definition of analytic continuation. Hence their derivatives also agree there, and therefore $G_{j-1} = G_j$ on the overlap. Since $G_{j-1} \equiv 0$, we have $G_j = 0$ on a non-empty open subset of D_j . By the identity theorem, $G_j \equiv 0$ in D_j .

By induction, the relation holds for every analytic element obtained along the path. $\square$

In particular, this result implies that if an analytic function satisfies a differential equation in a domain, then any analytic continuation of the function also satisfies the same differential equation in the extended domain. This result shows the uniqueness property of analytic continuation that once an analytic function satisfies a given functional relation locally, the relation continues to hold wherever the function is analytically continued.

Singularities associated with differential equations may be divided into two categories: fixed and movable. A fixed singularity is one whose position is prescribed by the differential equation and does not depend on the choice of solution. A movable singularity, however, arises from the solution itself and therefore depends on the values of integration constants or other internal parameters. As these parameters vary, the corresponding singularity may move within the complex plane.

Definition 2.2.7 (Isolated singularity). Let f be analytic in a punctured neighbourhood of $z_0 \in \mathbb{C}$. If f fails to be analytic at z_0 , then z_0 is called an *isolated singularity* of f .

Isolated singularities can be classified into three types:

- A *removable singularity*, if f can be extended analytically to z_0 .
- A *pole*, if $|f(z)| \rightarrow \infty$ as $z \rightarrow z_0$.
- An *essential singularity*, if the singularity is neither removable nor a pole.

In complex analysis, certain functions are naturally multi-valued. A function is called multi-valued if a single input in the complex plane corresponds to several possible values of the function. Examples are the complex logarithm $\log z$ and the square root $\sqrt{z}$, whose values depend on the path taken in the complex plane. To work with such functions analytically, we need to restrict the domain so that a single-valued branch of the function can be defined.

Definition 2.2.8 (Branch point). A point z_0 is called a *branch point* of a function if analytic continuation of the function around a closed loop encircling z_0 does not return to the original value of the function.

Definition 2.2.9 (Branch cut). A *branch cut* is a curve in the complex plane introduced to make a multi-valued function single-valued by restricting the domain so that analytic continuation around a branch point is no longer possible.

In the context of steady water waves, singularities of the analytically continued conformal mapping $z(\zeta)$ typically occur outside the unit circle in the ζ -plane. Tanveer [8] showed that these singularities are generically branch points of square-root type, and that their proximity to the unit circle controls both the regularity of the free surface and the convergence of series representations.

2.3 Implicit Function Theorem

The implicit function theorem is a fundamental result in analysis which provides conditions under which an equation involving several variables can be solved locally for one variable as a function of the others. In other words, it allows us to express a variable implicitly defined by an equation as a locally well-defined function, provided certain regularity and non-degeneracy conditions are satisfied.

Theorem 2.3.1 (Complex analytic Implicit Function Theorem). *Let $\mathcal{U} \subset \mathbb{C}^n \times \mathbb{C}^m$ be open, and let*

$$F : \mathcal{U} \rightarrow \mathbb{C}^m$$

be complex analytic. Write the variables as (z, w) , where $z \in \mathbb{C}^n$ and $w \in \mathbb{C}^m$. Suppose that $(z_0, w_0) \in \mathcal{U}$ satisfies

$$F(z_0, w_0) = 0,$$

and that the complex Jacobian matrix of F with respect to the variables w ,

$$D_w F(z_0, w_0) = \left(\frac{\partial F_i}{\partial w_j}(z_0, w_0) \right)_{1 \leq i, j \leq m},$$

is invertible.

Then there exist open neighbourhoods $X \subset \mathbb{C}^n$ of z_0 and $Y \subset \mathbb{C}^m$ of w_0 , with $X \times Y \subset \mathcal{U}$, and a unique complex analytic function

$$f : X \rightarrow Y$$

such that

$$f(z_0) = w_0$$

and

$$F(z, f(z)) = 0 \quad \text{for all } z \in X.$$

Moreover, every solution $(z, w) \in X \times Y$ of $F(z, w) = 0$ is of the form $w = f(z)$. The derivative of f is given by

$$Df(z) = -[D_w F(z, f(z))]^{-1} D_z F(z, f(z)), \quad z \in X.$$

The complex analytic implicit function theorem will be used later in Chapter 3 in the proof of Lewy's theorem. In that argument, one obtains an analytic relation between complex variables and needs to solve it locally for one variable as a function of the others. The theorem ensures that, provided the appropriate Jacobian is non-zero, this local solution is again complex analytic. This is one of the tools which allows analytic continuation across a real analytic free boundary.

Chapter 3

Analytic Continuation Across Free Boundaries

In many problems involving steady water waves, the fluid domain is not fixed in advance but is instead determined as part of the solution. The free surface of the fluid therefore acts as a boundary whose regularity and analytic properties play a central role in the mathematical structure of the problem. The natural question is whether harmonic or analytic functions defined inside the fluid can be continued across this free boundary. This question is closely connected to both the formation of singularities and the applicability of complex-analytic methods. The purpose of this chapter is to explain when and why such continuation is possible, and to describe a general mechanism that enforces analyticity across a boundary when suitable analytic boundary conditions are imposed.

For irrotational two-dimensional flows, the velocity potential and stream function are harmonic functions, and together they form a complex potential that is analytic inside the fluid domain. Since the velocity potential and stream function are harmonic conjugates, the complex potential $W(z) = \phi(x, y) + i\psi(x, y)$ is analytic in the interior of the fluid. However, analyticity across the free boundary is not automatic. In general, the free surface may obstruct continuation unless additional structure is present. In water wave problems, this additional structure is provided by the dynamic boundary condition, commonly given by Bernoulli's equation. When the boundary relation is analytic and non-degenerate, it can allow the harmonic functions to extend analytically across the free surface. This phenomenon explains why many steady wave profiles are real-analytic and also clarifies the mechanism by which singularities may form when continuation fails.

3.1 A general continuation principle

The basic mathematical result underlying analytic continuation across boundaries is proved by Lewy [7] using a theorem that applies to harmonic functions satisfying analytic boundary relations.

Before stating the theorem, we emphasize that the continuation is not imposed by symmetry,

but rather by the analytic structure of the boundary condition itself.

Theorem 3.1.1 (Lewy, 1952). *Let $U(x, y)$ be a real-valued harmonic function defined in a neighbourhood of the origin for $y < 0$, and let $V(x, y)$ be a harmonic conjugate of U , so that*

$$F(z) = U(x, y) + iV(x, y), \quad z = x + iy, \quad (3.1.1)$$

is analytic for $y < 0$. Suppose that U , V , $\partial U/\partial y$ and $\partial U/\partial x$ extend continuously to the boundary $y = 0$.

Assume that on $y = 0$ the normal derivative of U satisfies a boundary relation of the form

$$\frac{\partial U}{\partial y} = A\left(x, U, V, \frac{\partial U}{\partial x}\right), \quad (3.1.2)$$

where A is real-analytic in all its arguments.

Then U and V extend as real-analytic functions across the boundary $y = 0$ in a neighbourhood of the origin. Hence, $F = U + iV$ also extends as a complex analytic function.

This theorem states that an analytic boundary relation implies analytic continuation of harmonic conjugate pairs. The analyticity is not limited to the boundary itself, but holds in an open neighbourhood across the boundary.

Example: the case $A \equiv 0$ and the Schwarz reflection principle.

It is useful to compare Theorem 3.1.1 with the classical Schwarz reflection principle. The comparison is not exact at the level of the hypotheses, since Lewy's theorem is stated in terms of a differential boundary condition, whereas the Schwarz reflection principle is usually stated in terms of the boundary values of a holomorphic function. Nevertheless, the case $A \equiv 0$ shows that Lewy's theorem contains the Schwarz reflection principle as a special case.

Suppose, for example, that

$$F(z) = U(x, y) + iV(x, y), \quad z = x + iy,$$

is holomorphic for $y < 0$, extends continuously to $y = 0$, and has real boundary values on $y = 0$. Equivalently,

$$V(x, 0) = 0.$$

Then the Schwarz reflection principle gives an analytic extension to $y > 0$ by defining

$$\hat{F}(z) = \begin{cases} F(z), & y \leq 0, \\ \overline{F(\bar{z})}, & y > 0. \end{cases}$$

Indeed, on the boundary $y = 0$, one has

$$\overline{F(\bar{x})} = \overline{F(x)} = F(x),$$

because $F(x)$ is real. Hence the reflected function agrees with the original one on the common boundary.

We now explain how this fits into Lewy's theorem. Assume, in addition, the boundary regularity required in Theorem 3.1.1, so that the Cauchy–Riemann equations may be passed to the boundary. Since

$$V(x, 0) = 0,$$

we have

$$V_x(x, 0) = 0.$$

Using the Cauchy–Riemann relation

$$U_y = -V_x,$$

it follows that

$$U_y(x, 0) = 0.$$

Thus the boundary condition in Lewy's theorem becomes

$$U_y = A(x, U, V, U_x) \quad \text{with} \quad A \equiv 0.$$

Therefore, in this simple case, Lewy's theorem gives the same conclusion as the Schwarz reflection principle: the harmonic conjugate pair U, V , and hence the holomorphic function $F = U + iV$, extends analytically across the boundary $y = 0$.

The same argument also applies when V is constant, rather than zero, on the boundary. If

$$V(x, 0) = c_1$$

for some real constant c_1 , then $F - ic_1$ has real boundary values. Applying the Schwarz reflection principle to $F - ic_1$ gives the extension

$$\widehat{F}(z) = \begin{cases} F(z), & y \leq 0, \\ 2ic_1 + \overline{F(\bar{z})}, & y > 0. \end{cases}$$

Again $V_x(x, 0) = 0$, and hence $U_y(x, 0) = 0$, so this is again the case $A \equiv 0$ in Lewy's theorem.

Conversely, if $A \equiv 0$ in Lewy's boundary condition, then

$$U_y(x, 0) = 0.$$

By the Cauchy–Riemann equations,

$$V_x(x, 0) = -U_y(x, 0) = 0.$$

Hence V is constant along the boundary $y = 0$, at least locally. After subtracting this constant imaginary part, one obtains the usual Schwarz reflection situation. Thus the case $A \equiv 0$ may be viewed as the differential form of the Schwarz reflection principle.

This example shows why Lewy's theorem can be regarded as a generalisation of Schwarz reflection. In the Schwarz reflection principle, analytic continuation is obtained from a symmetry condition on the boundary values. In Lewy's theorem, this symmetry is replaced by the more general analytic boundary relation

$$U_y = A(x, U, V, U_x).$$

When $A \equiv 0$, the classical reflection mechanism is recovered; for a general real-analytic function A , the same conclusion follows from the analytic structure of the boundary condition rather than from an explicit reflection symmetry.

3.2 Explanation of the proof

We now explain the proof of Lewy's continuation theorem. The aim is to show how the analytic boundary relation

$$U_y = A(x, U, V, U_x)$$

causes the harmonic conjugate pair U, V to extend analytically across the boundary $y = 0$.

Since U and V are harmonic conjugates in the lower half-plane, we define

$$F(z) = \frac{1}{2}(U(x, y) + iV(x, y)), \quad z = x + iy. \quad (3.2.1)$$

Then F is holomorphic for $y < 0$. The factor $1/2$ is only a normalization, but it is convenient because it allows us to write

$$U = F + \overline{F}, \quad V = i(\overline{F} - F)$$

on the boundary.

The idea of the proof is to construct a second function G , defined in a small neighbourhood in the lower half-plane, such that G is holomorphic for $y < 0$, is C^1 up to the boundary $y = 0$, and satisfies

$$G(x, 0) = \overline{F(x, 0)}. \quad (3.2.2)$$

If such a function G exists, then on the boundary we would have

$$F + G = U, \quad i(G - F) = V, \quad (F + G)_x = U_x.$$

Thus the boundary condition can be written in terms of F and G . The proof constructs G by first determining it on the vertical line $x = 0$, and then extending it along horizontal lines $y = \text{constant}$. The construction is arranged so that G satisfies the differential equations that would necessarily hold if G were the reflected conjugate branch of F .

We first determine what equation G should satisfy on the boundary. Suppose for the moment that $G = \overline{F}$ on $y = 0$. Then

$$F + G = U, \quad i(G - F) = V.$$

Moreover, since

$$2F_x = U_x + iV_x, \quad 2G_x = U_x - iV_x,$$

we obtain

$$G_x - F_x = -iV_x.$$

Using the Cauchy–Riemann relation $U_y = -V_x$, this becomes

$$G_x - F_x = iU_y. \quad (3.2.3)$$

On the boundary, the hypothesis gives

$$U_y = A(x, U, V, U_x).$$

Substituting

$$U = F + G, \quad V = i(G - F), \quad U_x = (F + G)_x,$$

into the boundary condition, equation (3.2.3) becomes

$$G_x = F_x + iA(x, F + G, i(G - F), (F + G)_x). \quad (3.2.4)$$

This is the equation that G must satisfy on $y = 0$ if it is to play the role of $\bar{F}$ on the boundary. We therefore use this identity to motivate the differential equation imposed on G in the lower half-plane:

$$(E_x) \quad G_x = F_x + iA(z, F + G, i(G - F), (F + G)_x), \quad z = x + iy. \quad (3.2.5)$$

This equation will be used later to define G along horizontal lines.

However, we cannot construct G directly from (E_x) near the origin because the value of G on the line $x = 0$ is not yet known. Therefore we first define G along the vertical line $x = 0$, for $y \leq 0$.

To obtain the corresponding equation on $x = 0$, we use the holomorphic relations that should hold for F and G . Since F is holomorphic,

$$F_x = -iF_y.$$

If G is to be holomorphic, then similarly

$$G_x = -iG_y.$$

Substituting these into (E_x) , and using

$$(F + G)_x = -i(F_y + G_y),$$

we obtain

$$(E_y) \quad G_y(0, y) = F_y(0, y) - A\left(iy, F(0, y) + G(0, y), i(G(0, y) - F(0, y)), -i(F_y(0, y) + G_y(0, y))\right). \quad (3.2.6)$$

Thus (E_y) is the vertical version of the same equation, rewritten using the holomorphic relation between x - and y -derivatives.

We now prescribe the initial value of G at the origin. Since we want $G = \overline{F}$ on the boundary, we set

$$G(0, 0) = \overline{F(0, 0)} = \frac{1}{2}(U(0, 0) - iV(0, 0)). \quad (3.2.7)$$

We must also identify the initial value of $G_y(0, 0)$, because the equation (E_y) is implicit in G_y . If G is to be holomorphic and satisfy $G = \overline{F}$ on $y = 0$, then on the boundary

$$G_x(0, 0) = \frac{1}{2}(U_x(0, 0) - iV_x(0, 0)).$$

Since a holomorphic function satisfies $G_y = iG_x$, we define

$$G_y(0, 0) = \frac{i}{2}(U_x(0, 0) - iV_x(0, 0)). \quad (3.2.8)$$

Using $V_x = -U_y$, this can also be written as

$$G_y(0, 0) = -F_y(0, 0) + iU_x(0, 0) = F_y(0, 0) - U_y(0, 0). \quad (3.2.9)$$

This choice is exactly the one required for consistency with the boundary relation.

We now explain why (E_y) determines G_y locally. Write

$$p = G_y(0, y).$$

Then (E_y) has the form

$$p = F_y(0, y) - A\left(iy, F(0, y) + G, i(G - F(0, y)), -i(F_y(0, y) + p)\right).$$

Equivalently, define

$$\phi(y, G, p) = p - F_y(0, y) + A\left(iy, F(0, y) + G, i(G - F(0, y)), -i(F_y(0, y) + p)\right). \quad (3.2.10)$$

Then (E_y) is equivalent to

$$\phi(y, G, p) = 0.$$

At the initial point

$$y = 0, \quad G = G(0, 0), \quad p = G_y(0, 0),$$

the choice of $G(0, 0)$ and $G_y(0, 0)$, together with the boundary relation $U_y = A(x, U, V, U_x)$, gives

$$\phi(0, G(0, 0), G_y(0, 0)) = 0.$$

Thus we have a starting solution for the implicit equation.

To apply the complex implicit function theorem, we compute the derivative of ϕ with respect to p :

$$\frac{\partial \phi}{\partial p} = 1 - i A_4\left(iy, F + G, i(G - F), -i(F_y + p)\right), \quad (3.2.11)$$

where A_4 denotes the derivative of A with respect to its fourth argument. At the origin the arguments of A are real boundary values, and A_4 is real there. Hence the second term in (3.2.11) is purely imaginary, while the first term is 1. Therefore

$$\frac{\partial \phi}{\partial p}(0, G(0, 0), G_y(0, 0)) \neq 0.$$

The complex implicit function theorem therefore applies. Hence, near the origin, (E_y) can be solved uniquely for $p = G_y$ as an analytic function of y and G . In other words, (E_y) becomes an ordinary differential equation of the form

$$G_y(0, y) = B(y, G(0, y)), \quad (3.2.12)$$

for some analytic function B . By the classical existence and uniqueness theorem for ordinary differential equations, this determines $G(0, y)$ uniquely for $y \leq 0$ sufficiently close to 0, with initial value (3.2.7).

Having constructed G on the vertical line $x = 0$, we next extend G to nearby points with $x \neq 0$. For each fixed $y \leq 0$, we solve the horizontal equation (E_x) , taking the previously constructed value $G(0, y)$ as initial data. Thus, for each fixed y , we solve

$$G_x(x, y) = F_x(x, y) + iA(x + iy, F(x, y) + G(x, y), i(G(x, y) - F(x, y)), F_x(x, y) + G_x(x, y)). \quad (3.2.13)$$

This equation is also implicit in G_x , but the same implicit-function argument applies. Indeed, if $q = G_x$, then the equation has the form

$$q - F_x - iA(z, F + G, i(G - F), F_x + q) = 0.$$

The derivative with respect to q is again of the form

$$1 - iA_4,$$

which is nonzero near the origin by the same argument as above. Therefore (E_x) is a genuine ordinary differential equation in the variable x , depending continuously and analytically on the parameter y . Solving this equation with initial value $G(0, y)$ defines $G(x, y)$ in a small lower half-neighbourhood of the origin.

The two steps may therefore be summarized as follows. First, we solve (E_y) on the vertical line $x = 0$, using the initial values (3.2.7) and (3.2.8). This gives $G(0, y)$ for $y \leq 0$ near 0. Second, for each such y , we solve (E_x) along the horizontal line $y = \text{constant}$, using $G(0, y)$ as initial data. This gives $G(x, y)$ for $y \leq 0$ near the origin.

It remains to justify that the function G constructed in this way is holomorphic for $y < 0$. For this purpose, introduce the single complex differential equation

$$(E) \quad \frac{dG}{dz} = \frac{dF}{dz} + iA\left(z, F + G, i(G - F), \frac{dF}{dz} + \frac{dG}{dz}\right). \quad (3.2.14)$$

This is the complex form of (E_x) . Indeed, if F and G are holomorphic, then

$$\frac{dF}{dz} = F_x = -iF_y, \quad \frac{dG}{dz} = G_x = -iG_y.$$

Thus (E) reduces to (E_x) along horizontal lines, and to (E_y) along the vertical line $x = 0$.

Since (E) is implicit in dG/dz , we again use the implicit function theorem to solve for dG/dz . Choose a point iy_0 with $y_0 < 0$ inside the constructed lower-half-plane neighbourhood. At this point we already know the value $G(0, y_0)$. We also know the derivative that the complex equation must have there, namely

$$\left. \frac{dG}{dz} \right|_{z=iy_0} = -iG_y(0, y_0) = G_x(0, y_0). \quad (3.2.15)$$

This derivative is included because the differential equation (E) is implicit in dG/dz ; in order to apply the implicit function theorem, we must start from a point where the equation is already satisfied, including the derivative variable.

The implicit function theorem then rewrites (E) locally as an explicit holomorphic differential equation

$$\frac{dG}{dz} = \mathcal{B}(z, G),$$

where $\mathcal{B}$ is analytic. Standard existence and uniqueness theory for analytic ordinary differential equations gives a unique holomorphic solution near iy_0 . On the other hand, when this holomorphic solution is restricted to the vertical line, it satisfies the same equation (E_y) with the same initial value as the previously constructed function. When it is restricted to horizontal lines, it satisfies the same equation (E_x) with the same initial values. By uniqueness of the ordinary differential equations, the holomorphic solution of (E) must coincide with the function $G(x, y)$ constructed above. Therefore G is holomorphic in the interior of the lower-half-plane neighbourhood.

We now identify the boundary value of G . Since the x -axis is part of the boundary of the constructed region, the holomorphicity of G does not by itself give analyticity across $y = 0$. However, the hypotheses imply that F and F_x are continuous up to $y = 0$. Therefore the horizontal equation (E_x) depends continuously on the parameter y , and the initial values $G(0, y)$ also depend continuously on y . By the continuous dependence theorem for ordinary differential equations, the solutions $G(x, y)$ converge uniformly, as $y \rightarrow 0^-$, to a boundary function $G(x, 0)$.

The limiting function $G(x, 0)$ solves (E_x) on $y = 0$ with initial value

$$G(0, 0) = \overline{F(0, 0)}.$$

We claim that another solution of the same equation with the same initial value is

$$G(x, 0) = \overline{F(x, 0)}.$$

Indeed, if $G = \overline{F}$ on $y = 0$, then

$$F + G = U, \quad i(G - F) = V, \quad (F + G)_x = U_x.$$

Moreover,

$$G_x - F_x = (\overline{F})_x - F_x = -iV_x = iU_y.$$

Using the boundary relation $U_y = A(x, U, V, U_x)$, we obtain

$$G_x = F_x + iA(x, U, V, U_x).$$

Substituting

$$U = F + G, \quad V = i(G - F), \quad U_x = (F + G)_x,$$

shows that $G = \overline{F}$ satisfies exactly (E_x) on the boundary. Since the ordinary differential equation has a unique solution with the prescribed initial value, it follows that

$$G(x, 0) = \overline{F(x, 0)} \tag{3.2.16}$$

for x sufficiently close to 0.

Finally, we define an extension of F to the upper half-plane by reflection:

$$\tilde{F}(x + iy) = \overline{G(x - iy)}, \quad y \geq 0. \tag{3.2.17}$$

For $y < 0$, we keep the original definition

$$\tilde{F}(z) = F(z).$$

The function $\tilde{F}$ is holomorphic for $y < 0$, because it agrees there with the original F . It is also holomorphic for $y > 0$. Indeed, if G has a local power series expansion

$$G(w) = \sum_{n=0}^{\infty} a_n (w - w_0)^n,$$

then

$$\overline{G(\bar{z})} = \sum_{n=0}^{\infty} \overline{a_n} (z - \overline{w_0})^n,$$

which is holomorphic in z .

On the boundary $y = 0$, equation (3.2.16) gives

$$\overline{G(x, 0)} = F(x, 0).$$

Therefore the upper-half-plane definition agrees continuously with the original lower-half-plane definition on the boundary. Hence $\tilde{F}$ is continuous across $y = 0$, holomorphic for $y < 0$, and holomorphic for $y > 0$.

Now we only need to conclude that $\tilde{F}$ is holomorphic also on the boundary. This follows from Morera's theorem. Indeed, the function $\tilde{F}$ is continuous in a neighbourhood of the origin, and it is holomorphic away from the line $y = 0$. If a small triangle crosses the boundary, it can be divided into pieces lying in the upper and lower half-planes. The integrals over the interior boundary segments cancel because the function is continuous across $y = 0$, while the integrals over the pieces vanish by holomorphicity in each half-plane. Hence the integral of $\tilde{F}$ over every sufficiently small triangle is zero. By Morera's theorem, $\tilde{F}$ is holomorphic in a full

neighbourhood of the origin.

Thus F extends holomorphically across $y = 0$. Since

$$2F = U + iV,$$

the real and imaginary parts U and V extend as real-analytic functions across the boundary. This proves Lewy's continuation theorem.

3.3 Application to steady irrotational water waves

We will now see how Lewy's theorem applies to the steady two-dimensional irrotational water wave problem. The key point is that the dynamic boundary condition on the free surface leads to an analytic boundary relation of the type required by the theorem.

Consider a two-dimensional inviscid, incompressible, irrotational flow. Let $\phi(x, y)$ denote the velocity potential and $\psi(x, y)$ the stream function, so that

$$u = \nabla\phi, \quad \Delta\phi = 0, \tag{3.3.1}$$

and ψ is a harmonic conjugate of ϕ . The complex potential

$$W(z) = \phi(x, y) + i\psi(x, y), \quad z = x + iy, \tag{3.3.2}$$

is therefore analytic in the fluid domain.

We consider a steady travelling wave in a frame moving with the wave speed. In this frame, the free surface is a streamline, and we may choose coordinates so that

$$\psi = 0, \tag{3.3.3}$$

on the free surface, while the fluid occupies the region $\psi < 0$.

The dynamic boundary condition on the free surface is given by Bernoulli's equation. For a steady flow with gravity and without surface tension, this may be written in the form

$$\frac{1}{2}|\nabla\phi|^2 + gy = \text{constant}. \tag{3.3.4}$$

Since the constant in Bernoulli's equation can be absorbed into the definition of the vertical coordinate, we may rescale variables (following Lewy) so that the boundary condition takes the form,

$$T + y = 0 \quad \text{on } \psi = 0, \tag{3.3.5}$$

where $T = \frac{1}{2}|\nabla\phi|^2$ is the kinetic energy per unit mass.

We now regard the physical coordinates (x, y) as functions of (ϕ, ψ) . Since $z(W)$ is analytic for irrotational flow, both $x(\phi, \psi)$ and $y(\phi, \psi)$ are harmonic functions in the plane.

On the free surface $\psi = 0$, Bernoulli's equation can be rewritten in terms of $z(W)$ as

$$T = \frac{1}{2} \left| \frac{dz}{dW} \right|^{-2}. \quad (3.3.6)$$

Combining this with the boundary condition $T + y = 0$ yields

$$-2y \left| \frac{dz}{dW} \right|^2 = 1 \quad \text{on } \psi = 0. \quad (3.3.7)$$

Writing this equation in terms of the real and imaginary parts of $z(W)$ gives

$$-2y \left[\left(\frac{\partial y}{\partial \phi} \right)^2 + \left(\frac{\partial y}{\partial \psi} \right)^2 \right] = 1 \quad \text{on } \psi = 0. \quad (3.3.8)$$

This equation provides an analytic boundary relation involving the harmonic function $y(\phi, \psi)$ and its first derivatives along the boundary $\psi = 0$. Moreover, as long as $y < 0$ on the free surface, the coefficient multiplying $(\partial y / \partial \psi)^2$ is nonzero. Consequently, the boundary condition can be solved locally for the normal derivative $\partial y / \partial \psi$ as an analytic function of ϕ , y , and the tangential derivative $\partial y / \partial \phi$. This allows us to apply the real analytic implicit function theorem, which guarantees that the normal derivative $\partial y / \partial \psi$ can be expressed locally as an analytic function of ϕ , y , and $\partial y / \partial \phi$.

Thus Bernoulli's equation prescribes the normal derivative of a harmonic function through an analytic boundary relation. This is the structural assumption required by Lewy's theorem, and therefore the hypotheses of the theorem are satisfied.

Therefore, the hypotheses of Lewy's theorem are satisfied, and we conclude that $y(\phi, \psi)$ extends as a real-analytic function across the boundary $\psi = 0$ in a neighbourhood of any point where the fluid velocity is nonzero. Since $x(\phi, \psi)$ is harmonic conjugate to $y(\phi, \psi)$, it also extends analytically across the boundary.

Thus, away from stagnation points, the free surface parametrization $(x(\phi, 0), y(\phi, 0))$ is locally real-analytic. This result provides a rigorous justification for the analyticity of steady irrotational water wave profiles and explains why complex-analytic continuation plays a central role in their mathematical analysis.

The analytic continuation guaranteed by Lewy's theorem relies on a non-degeneracy condition. In the water wave context, this condition fails precisely at stagnation points, where the fluid velocity vanishes.

At such points, the boundary relation can no longer be solved analytically for the normal derivative. As a result, the continuation mechanism breaks down, allowing singularities such as corners or cusps to form on the free surface. This observation provides a mathematical explanation for the appearance of sharp-crested waves and other singular structures.

3.4 Continuation beyond the physical fluid domain

Lewy's theorem guarantees analytic continuation across the boundary $\psi = 0$ independently of the physical interpretation of the extended region. The continuation is determined solely by analyticity and uniqueness and does not rely on the existence of fluid motion beyond the free surface.

As a result, the analytic extension of $x(\phi, \psi)$ and $y(\phi, \psi)$ generally enters a region that has no direct physical meaning. Nevertheless, this unphysical domain plays an essential mathematical role: it encodes information about the geometry of the free surface and allows singularities of the physical solution to be studied through complex-analytic methods.

3.5 Analyticity of Capillary waves

In the absence of surface tension, Lewy proved that if a steady gravity wave solution is sufficiently regular near a point on the free surface, then the free surface is automatically real analytic in a neighbourhood of that point. A similar analyticity result holds when surface tension is included.

For a steady irrotational incompressible fluid with surface tension, the velocity field can be written as $u = \nabla\phi$, where the velocity potential ϕ satisfies Laplace's equation in the fluid domain

$$\Delta\phi = 0 \quad \text{in } \Omega_\eta. \quad (3.5.1)$$

On the free surface $y = \eta(x)$, the Bernoulli condition is modified to include the surface tension term involving the curvature κ ,

$$\frac{1}{2}|\nabla\phi|^2 + g\eta + \sigma\kappa = C \quad (3.5.2)$$

where $\sigma > 0$ denotes the surface tension coefficient.

In the presence of surface tension, similar analyticity results can be obtained. In the two-dimensional setting considered in this thesis, such results follow from the work of Matei [10], which extends Lewy's theorem to free boundary problems with surface tension. In particular, it is shown that under suitable regularity assumptions, the free surface and the associated harmonic functions are real analytic in a neighbourhood of the boundary.

More generally, Matei considered free-boundary problems of the form

$$\Delta\phi = 0 \quad \text{in } \Omega_\eta, \quad \mathbf{n} \cdot \nabla\phi = 0 \quad \text{on } y = \eta(x), \quad G(x, \nabla\phi) + \kappa = 0 \quad \text{on } y = \eta(x), \quad (3.5.3)$$

and proved that if the coefficients are sufficiently regular, then solutions automatically gain regularity and become analytic. Thus, as in the case of gravity waves, capillary waves also possess analytic free surfaces under suitable smoothness assumptions. The presence of the curvature term associated with surface tension leads to an elliptic structure that enforces this analyticity.

Chapter 4

Singularities in gravity steady water waves

Everything discussed in this chapter is based on the Tanveer's paper [8].

Equations, specifically the gravity type water waves have singularities in the analytically continued functions in the unphysical domain when there are no singularities in the physical domain. If a function is analytic inside a closed contour, it can be extended beyond the contour into a larger domain, unless we find a singularity beyond it.

Considering the two dimensional periodic deep water wave of period 2π , we will assume that the wave is stationary and is flowing to the right at a constant speed c . More assumptions are that the acceleration due to gravity in the negative y direction g is 1 and that the waves are symmetric, which means that the origin is chosen in such a way that the waves mirror each other across the y -axis. This would imply that the origin is either at the crest or trough. In this paper, the origin is assumed to coincide with the trough.

We will introduce a conformal map $z(\zeta)$ from a cut unit disk in the ζ -plane to a semi-infinite strip as shown in Figure 4.1b. In the disk, $\zeta = 1, 0$ and -1 (on each side of the cut) correspond to points A, B and D in the strip respectively (Figure: 4.1a). The free boundary of the unit disk is $|\zeta| = 1$ which is the circular boundary of the disk. Since in this paper, the origin ($\zeta = 0$) is assumed to coincide with the trough, then $\zeta = 1, -1$ are at crests.

The conformal mapping is represented as $z(\zeta) = 2\pi + i \ln \zeta + i f(\zeta) = 2\pi + i(\ln \zeta + f(\zeta))$ where $f(\zeta)$ is an analytic function. The paper mentions that on the real ζ axis, on the interval $(-1, 1)$, which is ray CD in the Figure 4.1a, it can be concluded that $\text{Im } f = 0$ using wave symmetry. We can prove this by the following:

We know that the conformal mapping is $z(\zeta) = 2\pi + i(\ln \zeta + f(\zeta))$. And also that $\ln \zeta = \ln |\zeta| + i \arg \zeta$. So,

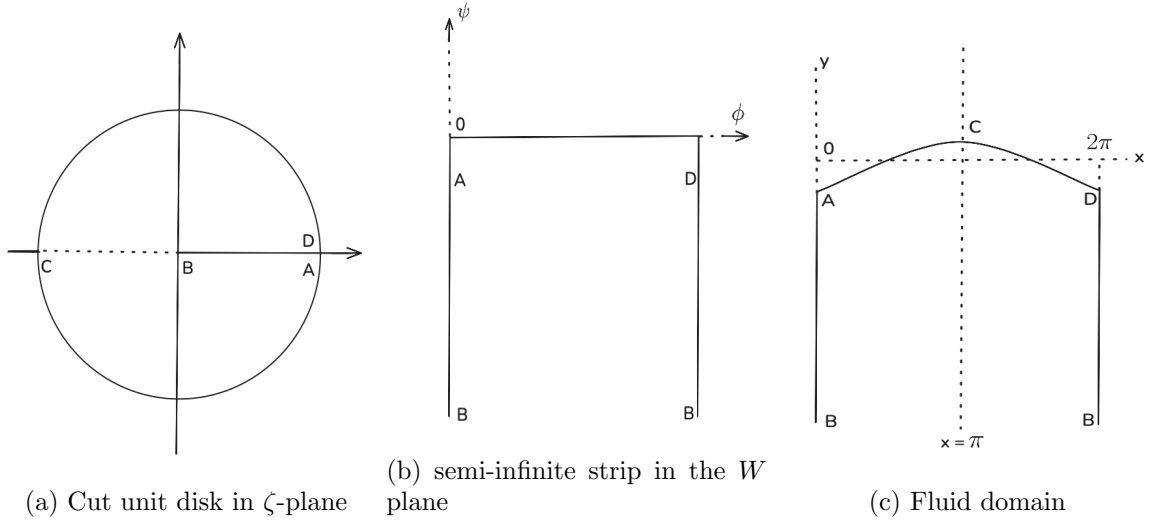


Figure 4.1

$$z(\zeta) = 2\pi + i \ln \zeta + i f(\zeta) \quad (4.0.1)$$

$$= 2\pi + i(\ln |\zeta| + i \arg \zeta) + i(\operatorname{Re} f(\zeta) + i \operatorname{Im} f(\zeta)) \quad (4.0.2)$$

$$= 2\pi - \arg \zeta - \operatorname{Im} f(\zeta) + i(\ln |\zeta| + \operatorname{Re} f(\zeta)). \quad (4.0.3)$$

Geometrically from the Figure 4.1, this conformal mapping maps the ray DB in the ζ -plane to the vertical ray DB in the z -plane, the ray CB in the ζ -plane to the vertical ray CB in the z -plane and the ray AB in the ζ -plane to the vertical ray AB in the z -plane. We claim that this mapping is possible if and only if $\operatorname{Im} f(\zeta) = 0$ when $\operatorname{Im} \zeta = 0$.

Now this is because if we take the vertical ray DB, it is given by $\operatorname{Re} z = 2\pi$. So, we need $2\pi - \arg \zeta - \operatorname{Im} f(\zeta) = 0$. But in the ζ -plane, $\arg \zeta = 0$ along DB. Thus, we get $2\pi - \operatorname{Im} f(\zeta) = 2\pi$ which is equivalent to saying that $\operatorname{Im} f(\zeta) = 0$. We can similarly prove this for the other rays.

So, $z(\bar{\zeta})$ and $z(\zeta)$ symmetric about $x = \pi$ due to the wave symmetry as shown in Figure 4.1c.

$$z(\bar{\zeta}) = 2\pi - x(\zeta) + iy(\zeta) = 2\pi - \overline{z(\zeta)}. \quad (4.0.4)$$

Now,

$$\operatorname{Im} z(\bar{\zeta}) = \operatorname{Im} z(\zeta) \quad (4.0.5)$$

$$\operatorname{Re} z(\bar{\zeta}) = 2\pi - \operatorname{Re} z(\zeta) \quad (4.0.6)$$

$$\arg \bar{\zeta} = 2\pi - \arg \zeta. \quad (4.0.7)$$

So, $z(\bar{\zeta})$ can be written as,

$$z(\bar{\zeta}) = 2\pi - \arg \bar{\zeta} - \operatorname{Im} f(\bar{\zeta}) + i|\ln |\bar{\zeta}| + \operatorname{Re} f(\bar{\zeta})| \quad (4.0.8)$$

$$= \arg \zeta - \operatorname{Im} f(\bar{\zeta}) + i|\ln |\zeta| + \operatorname{Re} f(\bar{\zeta})|. \quad (\text{using (4.0.7)}) \quad (4.0.9)$$

Here,

$$\operatorname{Re} z(\bar{\zeta}) = \arg \zeta - \operatorname{Im} f(\bar{\zeta}) \quad (4.0.10)$$

$$\operatorname{Im} f(\bar{\zeta}) = |\ln |\bar{\zeta}| + \operatorname{Re} f(\bar{\zeta})|. \quad (4.0.11)$$

Now $\operatorname{Im} z(\bar{\zeta}) = \operatorname{Im} z(\zeta)$ implies $\operatorname{Re} f(\bar{\zeta}) = \operatorname{Re} f(\zeta)$.

So we can derive that,

$$\operatorname{Re} z(\bar{\zeta}) = 2\pi - \operatorname{Re} z(\zeta) \implies \arg \zeta - \ln f(\bar{\zeta}) \quad (4.0.12)$$

$$= 2\pi - (2\pi - \arg \zeta - \operatorname{Im} f(\zeta)) \quad (4.0.13)$$

$$= \arg \zeta + \operatorname{Im} f(\zeta). \quad (4.0.14)$$

Thus, $\operatorname{Im} f(\zeta) = -\operatorname{Im} f(\bar{\zeta})$.

With this we can conclude that,

$$\begin{cases} \operatorname{Re} f(\zeta) = \operatorname{Re} f(\bar{\zeta}) \\ \operatorname{Im} f(\zeta) = -\operatorname{Im} f(\bar{\zeta}). \end{cases} \quad (4.0.15)$$

Now, on the real ζ -axis, $\zeta = \bar{\zeta}$. So,

$$\operatorname{Im} f(\zeta) = -\operatorname{Im} f(\zeta) \quad (4.0.16)$$

$$\Rightarrow \operatorname{Im} f(\zeta) = 0. \quad (4.0.17)$$

Thus, the analytic function $f(\zeta)$ is real since it's imaginary part is zero when $\zeta \in \mathbb{R}$. This analytic function which is analytic within the unit circle $|\zeta| \leq 1$ can be represented as a convergent power series:

$$f(\zeta) = \sum_{n=0}^{\infty} a_n \zeta^n. \quad (4.0.18)$$

Since we have proved that $\operatorname{Im} f(\zeta) = 0$, this would imply that a_n in the above convergent series are all real.

The radius of convergence of the power series in (4.0.18) is the distance from the origin to the nearest singularity of the function that it represents in the ζ -plane.

As previously mentioned, $f(\zeta)$ is analytic inside the unit circle and referring to Chapter 3, we consider waves with smooth profiles, $z(\zeta)$ and hence $f(\zeta)$ is analytic on the boundary of the unit circle, $|\zeta| = 1$. Since $z(\zeta) = 2\pi + i \ln \zeta + i f(\zeta)$ and the $\ln \zeta$ represents the branch cut, $f(\zeta)$ being analytic on $|\zeta| = 1$ ensures that the interface is smooth. If there were singularities on $|\zeta| = 1$, it would mean that there are non-smooth features like cusps on the boundary.

Thus, for $f(\zeta)$, the singularities all lie outside the unit circle, $|\zeta| > 1$. This would mean that the nearest singularity of $f(\zeta)$ is outside the unit circle at $\zeta = \zeta_0$. In reference to complex

analysis, the radius of convergence is greater than 1.

In the complex plane, the complex velocity potential is represented as $W = \phi + i\psi$ where $\phi(x, y)$ is the velocity potential and ψ is the stream function in the z -plane. In the paper, the mapping from the domain in the ζ -plane to the W -plane where $W = ic \ln \zeta + 2\pi c$. (Figure 4.1b).

Now consider the two mappings

$$z(\zeta) = 2\pi + i \ln \zeta + if(\zeta), \quad W(\zeta) = ic \ln \zeta + 2\pi c. \quad (4.0.19)$$

The complex velocity in the physical plane is given by

$$\frac{dW}{dz}. \quad (4.0.20)$$

Using the chain rule we may write

$$\frac{dW}{dz} = \frac{dW/d\zeta}{dz/d\zeta}. \quad (4.0.21)$$

Since the derivative $dz/d\zeta$ involves the term $f'(\zeta)$, any singularities of the analytic function $f(\zeta)$ appear in the complex velocity through this expression. We therefore compute the derivatives $dW/d\zeta$ and $dz/d\zeta$.

Since,

$$\frac{dW}{d\zeta} = \frac{ic}{\zeta} \quad \text{and} \quad \frac{dz}{d\zeta} = \frac{i}{\zeta} + if'(\zeta), \quad (4.0.22)$$

$$\frac{dW}{dz} = \frac{dW}{d\zeta} \cdot \frac{d\zeta}{dz} = \frac{ic}{\zeta} \cdot \left(\frac{1}{\frac{i}{\zeta} + if'(\zeta)} \right) = \frac{c}{1 + \zeta f'(\zeta)}. \quad (4.0.23)$$

And hence,

$$|\nabla \phi|^2 = \left| \frac{dW}{dz} \right|^2 = \frac{c^2}{|1 + \zeta f'(\zeta)|^2}. \quad (4.0.24)$$

Applying this into the Bernoulli's equation on the free surface we get:

$$\frac{1}{2} \frac{c^2}{|1 + \zeta f'(\zeta)|^2} + y = \frac{1}{2} |\nabla \phi|^2 + y = \text{const.} \quad (4.0.25)$$

$$y = \text{const.} - \frac{1}{2} \frac{c^2}{|1 + \zeta f'(\zeta)|^2}. \quad (4.0.26)$$

Evaluating on $\zeta = e^{iv}$, $v \in [0, 2\pi]$ we get,

$$\text{Re } f = -\frac{1}{2} \frac{c^2}{|1 + \zeta f'(\zeta)|^2}, \quad (4.0.27)$$

where the constant has been eliminated by redefining f . And since $a_n \in \mathbb{R}$ for all n ,

$$\overline{f(\zeta)} = \sum_0^{\infty} a_n \overline{\zeta^n} = \sum_0^{\infty} a_n \overline{\zeta}^n = f(\overline{\zeta}). \quad (4.0.28)$$

And,

$$\overline{f'(\zeta)} = \sum_1^{\infty} n a_n \overline{\zeta^{n-1}} = \sum_1^{\infty} n a_n \overline{\zeta}^{n-1} = f'(\overline{\zeta}). \quad (4.0.29)$$

On the unit circle, $\overline{\zeta} = \frac{1}{\zeta}$ and therefore,

$$\overline{f(\zeta)} = f(\overline{\zeta}) = f\left(\frac{1}{\zeta}\right) \quad (4.0.30)$$

$$\text{and, } \overline{f'(\zeta)} = f'(\overline{\zeta}) = f'\left(\frac{1}{\zeta}\right). \quad (4.0.31)$$

Now, the real part of the analytic function $f(\zeta)$ is,

$$\operatorname{Re} f(\zeta) = \frac{f(\zeta) + \overline{f(\zeta)}}{2}. \quad (4.0.32)$$

Using (4.0.30), this can be written on the unit circle as,

$$\operatorname{Re} f(\zeta) = \frac{f(\zeta) + f\left(\frac{1}{\zeta}\right)}{2}. \quad (4.0.33)$$

Using (4.0.27) we get,

$$f(\zeta) + f\left(\frac{1}{\zeta}\right) = \frac{-c^2}{|1 + \zeta f'(\zeta)|^2} = \left(-\frac{c^2}{(1 + \zeta f'(\zeta))(1 + \overline{\zeta f'(\zeta)})} \right). \quad (4.0.34)$$

Now the equation for $\operatorname{Re} f$, re-written as the above, can be analytically continued outside the unit disk. To do this, we represent,

$$F(\zeta) = f(\zeta) + f\left(\frac{1}{\zeta}\right), \quad (4.0.35)$$

and,

$$P(\zeta) = 1 + \frac{1}{\zeta} f'\left(\frac{1}{\zeta}\right). \quad (4.0.36)$$

So, (4.0.34) becomes,

$$F(\zeta) = -\frac{\beta}{(1 + \zeta f'(\zeta))(P(\zeta))}, \quad (\beta = c^2) \quad (4.0.37)$$

that is,

$$F(1 + \zeta f'(\zeta)) = -\frac{\beta}{P}. \quad (4.0.38)$$

We will now show that $\zeta F_\zeta + P = 1 + \zeta f'(\zeta)$.

First,

$$F_\zeta = \frac{d}{d\zeta} \left[f(\zeta) + f\left(\frac{1}{\zeta}\right) \right] = f'(\zeta) + f'\left(\frac{1}{\zeta}\right) \left(-\frac{1}{\zeta^2}\right) = f'(\zeta) - \frac{1}{\zeta^2} f'\left(\frac{1}{\zeta}\right), \quad (4.0.39)$$

which implies that,

$$\zeta F_\zeta = \zeta f'(\zeta) - \frac{1}{\zeta} f'\left(\frac{1}{\zeta}\right). \quad (4.0.40)$$

Using (4.0.40) and (4.0.36),

$$\zeta F_\zeta + P = \zeta f'(\zeta) - \frac{1}{\zeta} f'\left(\frac{1}{\zeta}\right) + 1 + \frac{1}{\zeta} f'\left(\frac{1}{\zeta}\right) = 1 + \zeta f'(\zeta). \quad (4.0.41)$$

Now using (4.0.41) in (4.0.38) we set,

$$\zeta F F_\zeta + P F = F(\zeta F_\zeta + P) = F(1 + \zeta f'(\zeta)) = -\frac{\beta}{P}. \quad (4.0.42)$$

Now we write $F_1 = F^{-1}$ and multiply both sides of the above equation by F^{-3} ,

$$\frac{\zeta F_\zeta}{F^2} + \frac{P}{F^2} = -\frac{\beta}{P F^3} \quad (4.0.43)$$

$$\Leftrightarrow -\frac{\zeta F_\zeta}{F^2} = \frac{P}{F^2} + \frac{\beta}{P F^3} \quad (4.0.44)$$

$$\Leftrightarrow \zeta F_1' = P F_1^2 + \frac{\beta}{P} F_1^3 \quad (4.0.45)$$

$$\Leftrightarrow F_1' = \frac{P}{\zeta} F_1^2 + \frac{\beta}{P \zeta} F_1^3. \quad (4.0.46)$$

The above equation is a non-linear first-order differential equation for F_1 . $P(\zeta) = 1 + \frac{1}{\zeta} f'\left(\frac{1}{\zeta}\right)$ and f' is analytic inside the unit disk, so P is analytic outside since $|\frac{1}{\zeta}| = \frac{1}{|\zeta|} < 1$ if $|\zeta| > 1$.

This implies that the coefficients of F_1^2 and F_1^3 are analytic outside the unit disk. If we assume that P is not unknown, then the above equation becomes a non-linear first-order differential equation. This is solved by Painleve [4]. This kind of non-linear ODE only allows movable singularities. That is the location of its branch points or poles, which are its singularities, depends on initial conditions on F .

For first-order equations of the form

$$w' = F(z, w) = \frac{\mathcal{P}(z, w)}{\mathcal{Q}(z, w)}, \quad (4.0.47)$$

where,

$$\mathcal{P}(z, w) = \sum_{k=0}^p \mathcal{P}_k(z) w^k \quad (4.0.48)$$

$$\mathcal{Q}(z, w) = \sum_{k=0}^q \mathcal{Q}_k(z) w^k \quad (4.0.49)$$

movable singularities are poles or algebraic branch points, and when the solution becomes unbounded one gets a local expansion in fractional powers of $z - z_1$. In particular, when the dominant balance leads to exponent $1/2$, the singularity is a square-root branch point.

For the equation

$$F_1' = \frac{P(\zeta)}{\zeta} F_1^2 + \frac{\beta}{P(\zeta)\zeta} F_1^3, \quad (4.0.50)$$

the key observation is that near a movable singularity $\zeta = \zeta_0$ at which $F_1 \rightarrow \infty$, the cubic term dominates the quadratic one.

The equation (4.0.50) has the form,

$$w' = A(\zeta) w^2 + B(\zeta) w^3, \quad (4.0.51)$$

with

$$A(\zeta) = \frac{P(\zeta)}{\zeta}, \quad B(\zeta) = \frac{\beta}{P(\zeta)\zeta}. \quad (4.0.52)$$

This is a first-order nonlinear differential equation polynomial in w . To determine the local behaviour of solutions near a movable singularity $\zeta = \zeta_0$, we apply Theorem 3.2.3 of Hille [4], which describes the dominant behaviour of solutions of differential equations of the form,

$$w' = \sum_{k=0}^p a_k(z) w^k, \quad (4.0.53)$$

where $w \rightarrow \infty$.

Theorem 4.0.1 (Hille, 1976). *Equation (4.0.53) has no movable infinitudes if $p < q + 2$. There are movable poles if $p = q + 2$. The poles are simple unless accidentally $P_p(z)$ vanishes at the point in question. If $p > q + 2$, there are movable branch points where the solution becomes infinite. Normally $p - q - 1$ branches are permuted at such a point.*

Note that Hille proves this theorem under stronger assumptions on $\mathcal{P}$ and $\mathcal{Q}$, but by the inspection of the proof, it shows that the result still applies in this setting.

The key result is, if the highest power term dominates, then near the singularity the solution behaves like $w(z) \sim C(z - z_0)^{-1/(p-1)}$. This gives the 'type' of singularity.

In the equation (4.0.50) the polynomial degree in F_1 is 3,

$$F_1' = A(\zeta) F_1^2 + B(\zeta) F_1^3. \quad (4.0.54)$$

The highest power is F_1^3 . So, $p = 3$.

The theorem also considers whether lower powers appear starting from w^q . The smallest power is F_1^2 . But when studying behaviour as $F_1 \rightarrow \infty$, the largest power dominates, so effectively the leading balance is $F_1' \sim B(\zeta)F_1^3$. Thus the dominant equation becomes $F_1' = B(\zeta_0)F_1^3$. This is why we set,

$$p = 3, \quad q = 0. \quad (4.0.55)$$

The cubic term controls the singular behaviour. The theorem gives the local exponent $\alpha = \frac{1}{p-1}$. Since $p = 3$ we get $\alpha = \frac{1}{2}$. According to Theorem 3.2.3, solutions then behave locally like $F_1(\zeta) \sim C(\zeta - \zeta_0)^{-1/(p-1)}$. With $p = 3$, therefore $F_1(\zeta) \sim C(\zeta - \zeta_0)^{-1/2}$.

Once the leading exponent is known, the solution admits an expansion near ζ_0 of the form,

$$F_1(\zeta) = \frac{1}{(\zeta - \zeta_0)^{1/2}} \sum_{n=0}^{\infty} \tilde{b}_n (\zeta - \zeta_0)^{n/2}. \quad (4.0.56)$$

with $\tilde{b}_0 \neq 0$. That is, the function F_1 is a series that approaches the neighbourhood of the movable singularities ζ_0 .

The expansion (4.0.56) contains half-integer powers of $\zeta - \zeta_0$, indicating that ζ_0 is a branch point of the analytically continued solution. We should just note that in the presence of such branch point singularities, analytic continuation in the complex plane may depend on the path along which the continuation is performed. In particular, continuation along different paths encircling the singularity may lead to different branches of the analytic function, so that the analytically continued solution may take different values depending on the path taken around the singular point. In this analysis, this reflects a multi-valued structure associated with the singularities of the conformal mapping in the complex ζ -plane.

We have previously set $F_1 = F^{-1}$. Now in equation (4.0.56), let $u = (\zeta - \zeta_0)^{1/2}$. So that $\zeta - \zeta_0 = u^2$. (4.0.56) can be written as,

$$F_1 = \frac{1}{u} \sum_{n=0}^{\infty} \tilde{b}_n u^n = u^{-1} \left(\tilde{b}_0 + \tilde{b}_1 u + \tilde{b}_2 u^2 + \dots \right). \quad (4.0.57)$$

Now,

$$F(\zeta) = \frac{1}{F_1(\zeta)} \quad (4.0.58)$$

$$= \frac{1}{\frac{1}{u} \left(\tilde{b}_0 + \tilde{b}_1 u + \tilde{b}_2 u^2 + \dots \right)} \quad (4.0.59)$$

$$= \frac{u}{\tilde{b}_0 + \tilde{b}_1 u + \tilde{b}_2 u^2 + \dots} \quad (4.0.60)$$

Thus, we get,

$$F(\zeta) = \sum_{n=1}^{\infty} b_n (\zeta - \zeta_0)^{\frac{n}{2}}, \quad (4.0.61)$$

using that $\frac{1}{b_0+z}$ is analytic for $|z| < |\tilde{b}_0|$. $f(1/\zeta)$ is analytic for $|\zeta| > 1$ so that any singularity of $F(\zeta)$ is a singularity of $f(\zeta)$ for $|\zeta| > 1$. From (4.0.35),

$$f(\zeta) = F(\zeta) - f(1/\zeta). \quad (4.0.62)$$

Substituting this into (4.0.61),

$$f(\zeta) = \sum_{n=1}^{\infty} b_n (\zeta - \zeta_0)^{\frac{n}{2}} - f(1/\zeta). \quad (4.0.63)$$

We have already established that $f(1/\zeta)$ is analytic at $\zeta = \zeta_0$. So, it can be expanded as a Taylor series around $\zeta = \zeta_0$,

$$f(1/\zeta) = \sum_{m=0}^{\infty} c_m (\zeta - \zeta_0)^m. \quad (4.0.64)$$

Thus,

$$f(\zeta) = F(\zeta) - f(1/\zeta) \quad (4.0.65)$$

$$f(\zeta) = \sum_{n=1}^{\infty} b_n (\zeta - \zeta_0)^{\frac{n}{2}} - f(1/\zeta) \quad (4.0.66)$$

$$f(\zeta) = \sum_{n=1}^{\infty} b_n (\zeta - \zeta_0)^{\frac{n}{2}} - \sum_{m=0}^{\infty} c_m (\zeta - \zeta_0)^m. \quad (4.0.67)$$

And this can thus be written as,

$$f(\zeta) = \sum_{n=0}^{\infty} d_n (\zeta - \zeta_0)^{\frac{n}{2}}. \quad (4.0.68)$$

Now looking at equations (4.0.67) and (4.0.68), we can see that if n is odd, then $d_n = b_n$. This is because the expansion of $f(1/\zeta)$ contains all the integer power terms leaving the ones whose powers cannot be integers but are half the odd integers.

Now let's see what happens to d_n when n is even. In equation (4.0.67), let $m = n/2$. So, $(\zeta - \zeta_0)^{\frac{n}{2}} = (\zeta - \zeta_0)^m$. So the coefficient of $(\zeta - \zeta_0)^m$ for $n/2 = m$ is $d_n = b_n - c_m$.

In the Taylor series of $f(1/\zeta)$, the coefficient c_m is,

$$c_m = \frac{1}{m!} \frac{d^m}{d\zeta^m} f(1/\zeta) \Big|_{\zeta=\zeta_0} \quad (4.0.69)$$

$$c_m = \frac{1}{(n/2)!} \frac{d^{n/2}}{d\zeta^{n/2}} f(1/\zeta) \Big|_{\zeta=\zeta_0} \quad (4.0.70)$$

$$d_n = b_n - \frac{1}{(n/2)!} \frac{d^{n/2}}{d\zeta^{n/2}} f(1/\zeta) \Big|_{\zeta=\zeta_0} \quad (4.0.71)$$

Next we study the behaviour of F at infinity. In order to do this, we define,

$$\zeta_1 = \frac{1}{\zeta}. \quad (4.0.72)$$

So, $\zeta \rightarrow \infty$ means that $\zeta_1 \rightarrow 0$

This implies that,

$$\frac{d\zeta_1}{d\zeta} = \frac{d}{d\zeta} \left(\frac{1}{\zeta} \right) = -\frac{1}{\zeta^2} = -\zeta_1^2. \quad (4.0.73)$$

Now using the chain rule,

$$F_\zeta = \frac{dF}{d\zeta} = \frac{dF}{d\zeta_1} \frac{d\zeta_1}{d\zeta} = \frac{dF}{d\zeta_1} \cdot (-\zeta_1^2) = -\zeta_1^2 \frac{dF}{d\zeta_1}. \quad (4.0.74)$$

Thus,

$$\zeta F F_\zeta = \left(\frac{1}{\zeta_1} \right) F (-\zeta_1^2 \frac{dF}{d\zeta_1}) = -\frac{\zeta_1^2}{\zeta_1} F \frac{dF}{d\zeta_1} = -\zeta_1 F \frac{dF}{d\zeta_1}. \quad (4.0.75)$$

Substituting the above in (4.0.42) we get,

$$-\zeta_1 F \frac{dF}{d\zeta_1} + P F = -\frac{\beta}{P} \quad (4.0.76)$$

where,

$$\frac{\beta}{P} = \frac{1}{1 + \frac{1}{\zeta} f'(\frac{1}{\zeta})} = \frac{\beta}{1 + \zeta_1 f'(\zeta_1)}. \quad (4.0.77)$$

Now, differentiating (4.0.18),

$$f'(\zeta) = \frac{d}{d\zeta} \sum_{n=0}^{\infty} a_n \zeta^n = \sum_{n=1}^{\infty} n a_n \zeta^{n-1}. \quad (4.0.78)$$

In order to compute $P(\zeta)$ in terms of $\zeta_1 = \frac{1}{\zeta}$ and expand it around $\zeta_1 = 0$, we have,

$$P = 1 + \zeta_1 f'(\zeta_1) \quad (4.0.79)$$

$$= 1 + \sum_{n=1}^{\infty} n a_n \zeta_1^n \quad (4.0.80)$$

$$= 1 + a_1 + 2a_2 \zeta_1^2 + 3a_3 \zeta_1^3 + \dots \quad (4.0.81)$$

In order to perform leading-order analysis, we substitute $P = 1$ into (4.0.76),

$$-\zeta_1 F \frac{dF}{d\zeta_1} + F = -\beta \quad (4.0.82)$$

$$\zeta_1 F \frac{dF}{d\zeta_1} = F + \beta \quad (4.0.83)$$

$$\zeta_1 \frac{dF}{d\zeta_1} = 1 + \frac{\beta}{F}. \quad (4.0.84)$$

This is a separable ODE in F and ζ_1 ,

$$\frac{dF}{1 + \frac{\beta}{F}} = \frac{d\zeta_1}{\zeta_1} \quad (4.0.85)$$

$$\int \frac{F}{F + \beta} dF = \int \frac{d\zeta_1}{\zeta_1}. \quad (4.0.86)$$

If $u = F + \beta$, $du = dF$, $F = u - \beta$, then the LHS would be,

$$\int \frac{u - \beta}{u} du = \int \left(1 - \frac{\beta}{u}\right) du = u - \beta \ln u = F + \beta - \beta \ln(F + \beta). \quad (4.0.87)$$

Thus,

$$F + \beta - \beta \ln(F + \beta) = \int \frac{d\zeta_1}{\zeta_1} \quad (4.0.88)$$

$$\ln \zeta_1 = F + \beta - \beta \ln(F + \beta) - C. \quad (4.0.89)$$

Taking exponential on either side,

$$\zeta_1 = e^{-C} e^{F+\beta} (F + \beta)^{-\beta} = k e^F (F + \beta)^{-\beta} \quad (k = e^{-C} e^\beta). \quad (4.0.90)$$

In the leading-order approximation of F , The solution F is denoted by F_0 ,

$$\zeta_1 = k e^{F_0} (F_0 + \beta)^{-\beta}. \quad (4.0.91)$$

As $\zeta_1 \rightarrow 0$, or, $\zeta \rightarrow \infty$, we will derive the asymptotic form of F_0 ,

Take natural log on both sides of (4.0.91),

$$\ln \zeta_1 = \ln \left(k e^{F_0} (F_0 + \beta)^{-\beta} \right) \quad (4.0.92)$$

$$= \ln k + F_0 - \beta \ln(F_0 + \beta) \quad (4.0.93)$$

$$F_0 = -\ln \zeta - \ln k + \beta \ln(F_0 + \beta) \quad \left(\zeta_1 = \frac{1}{\zeta}\right). \quad (4.0.94)$$

As $\zeta \rightarrow \infty$, $\ln \zeta$ becomes quite large. So, we can write that,

$$F_0 \approx -\ln \zeta + \text{very small terms} \quad (4.0.95)$$

$$F_0 \approx -\ln \zeta + u. \quad (4.0.96)$$

where, $u \approx -\ln k + \beta \ln(F_0 + \beta)$

$$\beta + F_0 \approx \beta - \ln \zeta + u \quad (4.0.97)$$

$$\ln(\beta + F_0) \approx \ln(\beta - \ln \zeta + u). \quad (4.0.98)$$

So we can write $u \approx -\ln k + \beta \ln(F_0 + \beta)$ as,

$$u \approx -\ln k + \beta \ln(\beta - \ln \zeta + u). \quad (4.0.99)$$

We have already established that as $\zeta \rightarrow \infty$, $\ln \zeta$ becomes quite large. So,

$$\beta - \ln \zeta + u \approx -\ln \zeta \quad (4.0.100)$$

$$\ln(\beta - \ln \zeta + u) \approx \ln(-\ln \zeta). \quad (4.0.101)$$

Thus,

$$u \approx -\ln k + \beta \ln(-\ln \zeta). \quad (4.0.102)$$

Due to very large $\ln \zeta$, $\ln k$ becomes almost insignificant.

$$u \approx \beta \ln(-\ln \zeta). \quad (4.0.103)$$

Hence, $F_0 \approx -\ln \zeta + u$ becomes,

$$F_0 \approx -\ln \zeta + \beta \ln(-\ln \zeta). \quad (4.0.104)$$

In this chapter we investigated the analytic continuation of the conformal mapping formulation for steady periodic gravity waves and examined the resulting singularity structure in the complex ζ -plane. By analysing the functional relations obtained from Bernoulli's equation and exploiting the symmetry properties of the wave profile, the problem was reduced to a nonlinear first-order differential equation for the analytically continued function F .

The analysis shows that the singularities of the solution lie outside the unit circle and arise as movable singularities of the governing differential equation. Using the local behaviour of the solution near such points, it follows that these singularities exhibit square-root type branch behaviour. This provides a characterization of the analytic structure of steady water wave solutions in the complex plane.

The asymptotic analysis carried out near infinity further shows the behaviour of the analytically continued solution and supports the presence of these branch-point singularities. These results form the basis for understanding how complex singularities influence the structure of steady wave profiles and will be important in further analysis of the analytic continuation of water wave solutions.

Chapter 5

Global Analyticity in Capillary Waves

This chapter presents a formulation of two-dimensional free-surface Euler flows with surface tension based on global analyticity. The aim is to explain how the classical exact capillary-wave solutions discovered by Crapper arise naturally from analyticity constraints in the complex plane.

The analysis emphasizes analytic continuation, the structure of singularities of the conformal map, and the appearance of a Riccati-type equation derived from the Bernoulli boundary condition.

We consider steady periodic capillary waves of permanent form propagating on deep water in an ideal, incompressible, irrotational fluid with surface tension.

In order to describe periodic travelling waves, it is convenient to map the fluid domain conformally from the unit disc in the complex ζ -plane to the physical fluid region in the z -plane. Periodicity in the horizontal direction is incorporated through a logarithmic term in the mapping, since $\log \zeta$ increases by $2\pi i$ when ζ winds once around the unit circle.

For capillary waves of wavelength $2\pi/k$, observed in a frame moving with wave speed c , the conformal map is taken to be

$$z(\zeta) = \frac{2\pi}{k} + \frac{i}{k}(\log \zeta + f(\zeta)), \quad (5.0.1)$$

where f is analytic for $|\zeta| < 1$. The logarithmic term explicitly encodes the spatial periodicity, with the free surface corresponding to $|\zeta| = 1$.

Let the complex potential be

$$w(z) = \phi(x, y) + i\psi(x, y), \quad (5.0.2)$$

analytic throughout the fluid domain. In steady problems, the kinematic boundary condition implies

$$\psi = 0 \quad \text{on the free surface.} \quad (5.0.3)$$

The logarithmic dependence follows from the periodic geometry of the problem, while the constant term corresponds to the uniform background flow at infinity.

This constraint, together with uniform flow at infinity, determines the functional form $W(\zeta) = w(z(\zeta))$. For capillary waves, it is found uniquely that

$$W(\zeta) = \frac{2\pi c}{k} + \frac{ic}{k} \log \zeta. \quad (5.0.4)$$

On the free surface the non-dimensional Bernoulli condition takes the form,

$$\frac{1}{2}|w_z|^2 + \kappa = \text{const}, \quad (5.0.5)$$

where κ denotes curvature. The curvature formula is,

$$\kappa = \frac{\det(\gamma', \gamma'')}{\|\gamma'\|^3}, \quad (5.0.6)$$

Here $\gamma(\theta)$ denotes the parametrization of the free surface in the physical plane. The free surface is parametrized by

$$\gamma(\theta) = z(e^{i\theta}), \quad \zeta = e^{i\theta}, \quad |\zeta| = 1. \quad (5.0.7)$$

On the free surface $|\zeta| = 1$, let

$$\gamma(\theta) = z(\zeta), \quad \zeta = e^{i\theta}. \quad (5.0.8)$$

Then differentiation with respect to θ gives

$$\gamma'(\theta) = \frac{d}{d\theta} z(e^{i\theta}) = iz_\zeta(\zeta)\zeta, \quad (5.0.9)$$

and,

$$\gamma''(\theta) = \frac{d}{d\theta} (iz_\zeta(\zeta)) \quad (5.0.10)$$

$$= -\zeta z_\zeta(\zeta) - \zeta^2 z_{\zeta\zeta}(\zeta). \quad (5.0.11)$$

The speed of this parametrization is $\|\gamma'(\theta)\| = |\zeta z_\zeta(\zeta)|$. Since $|\zeta| = 1$, this may be written as

$$\|\gamma'(\theta)\|^2 = \zeta z_\zeta(\zeta) \overline{\zeta z_\zeta(\zeta)} \quad (5.0.12)$$

$$= \zeta z_\zeta(\zeta) \zeta^{-1} \overline{z_\zeta(\zeta)}, \quad (5.0.13)$$

so that

$$\|\gamma'(\theta)\| = (\zeta z_\zeta(\zeta) \zeta^{-1} \overline{z_\zeta(\zeta)})^{1/2}. \quad (5.0.14)$$

Now introduce,

$$A(\zeta) = \zeta z_\zeta(\zeta), \quad B(\zeta) = \zeta^{-1} \overline{z_\zeta(\zeta)}. \quad (5.0.15)$$

Then $\|\gamma'(\theta)\| = (AB)^{1/2}$. Instead of evaluating $\det(\gamma', \gamma'')$ directly in coordinates, it is convenient to use the equivalent complex form of the signed curvature:

$$\kappa = -\frac{i}{2\|\gamma'(\theta)\|} \frac{d}{d\theta} \log \left(\frac{\gamma'(\theta)}{\overline{\gamma'(\theta)}} \right). \quad (5.0.16)$$

Since

$$\gamma'(\theta) = iA(\zeta), \quad \overline{\gamma'(\theta)} = -iB(\zeta), \quad (5.0.17)$$

we obtain,

$$\frac{\gamma'(\theta)}{\overline{\gamma'(\theta)}} = -\frac{A(\zeta)}{B(\zeta)}. \quad (5.0.18)$$

The constant factor -1 disappears upon differentiation, hence

$$\kappa = -\frac{i}{2(AB)^{1/2}} \frac{d}{d\theta} \log \left(\frac{A}{B} \right). \quad (5.0.19)$$

Using $\frac{d}{d\theta} = i\zeta \frac{d}{d\zeta}$, this becomes

$$\kappa = \frac{\zeta}{2(AB)^{1/2}} \frac{d}{d\zeta} \log \left(\frac{A}{B} \right). \quad (5.0.20)$$

Now use the identity,

$$\frac{d}{d\zeta} \left(\frac{A}{B} \right)^{1/2} = \frac{1}{2} \left(\frac{A}{B} \right)^{1/2} \frac{d}{d\zeta} \log \left(\frac{A}{B} \right), \quad (5.0.21)$$

to rewrite the previous expression as

$$\kappa = \frac{\zeta}{A} \frac{d}{d\zeta} \left(\frac{A}{B} \right)^{1/2}. \quad (5.0.22)$$

Since $A = \zeta z_\zeta(\zeta)$, we have $\zeta/A = 1/z_\zeta(\zeta)$, and therefore,

$$\kappa = \frac{1}{z_\zeta(\zeta)} \frac{d}{d\zeta} \left(\frac{\zeta z_\zeta(\zeta)}{\zeta^{-1} \overline{z_\zeta(\zeta^{-1})}} \right)^{1/2}. \quad (5.0.23)$$

With the sign convention used in the chapter, this is written as

$$\kappa = -\frac{1}{z_\zeta(\zeta)} \frac{d}{d\zeta} \left(\frac{\zeta z_\zeta(\zeta)}{\zeta^{-1} \overline{z_\zeta(\zeta^{-1})}} \right)^{1/2}. \quad (5.0.24)$$

(The overall sign depends on the chosen orientation of the free-surface parametrisation and the sign convention for signed curvature.)

Next, since $W(\zeta) = w(z(\zeta))$, the chain rule gives $w_z = \frac{W_\zeta}{z_\zeta}$. Hence on $|\zeta| = 1$,

$$|w_z|^2 = \frac{W_\zeta(\zeta) \overline{W_\zeta(\zeta^{-1})}}{z_\zeta(\zeta) \overline{z_\zeta(\zeta^{-1})}}, \quad (5.0.25)$$

where we use the reflected boundary values on the unit circle. Substituting this and (5.0.24) into the Bernoulli condition,

$$\frac{1}{2}|w_z|^2 + \kappa = \text{const}, \quad (5.0.26)$$

gives

$$\frac{W_\zeta(\zeta) W_\zeta(\zeta^{-1})}{2 z_\zeta(\zeta) \bar{z}_\zeta(\zeta^{-1})} - \frac{1}{z_\zeta(\zeta)} \frac{d}{d\zeta} \left(\frac{\zeta z_\zeta(\zeta)}{\zeta^{-1} \bar{z}_\zeta(\zeta^{-1})} \right)^{1/2} = \text{const.} \quad (5.0.27)$$

Multiplying through by $z_\zeta(\zeta)$, and denoting the Bernoulli constant by γ , yields

$$\frac{W_\zeta(\zeta) W_\zeta(\zeta^{-1})}{2 \bar{z}_\zeta(\zeta^{-1})} - \frac{d}{d\zeta} \left(\frac{\zeta z_\zeta(\zeta)}{\zeta^{-1} \bar{z}_\zeta(\zeta^{-1})} \right)^{1/2} = \gamma z_\zeta(\zeta). \quad (5.0.28)$$

Equivalently,

$$\frac{W_\zeta(\zeta) W_\zeta(\zeta^{-1})}{2 z_\zeta(\zeta) \bar{z}_\zeta(\zeta^{-1})} = -\frac{d}{d\zeta} \left(\frac{\zeta z_\zeta(\zeta)}{\zeta^{-1} \bar{z}_\zeta(\zeta^{-1})} \right)^{1/2} + \gamma z_\zeta(\zeta), \quad (5.0.29)$$

Rewriting (5.0.5) in terms of ζ yields

$$\frac{W_\zeta(\zeta) W_\zeta(\zeta^{-1})}{2 \bar{z}_\zeta(\zeta^{-1})} = -\frac{d}{d\zeta} \left(\frac{\zeta z_\zeta(\zeta)}{\zeta^{-1} \bar{z}_\zeta(\zeta^{-1})} \right)^{1/2} + \gamma z_\zeta(\zeta), \quad (5.0.30)$$

where γ denotes the Bernoulli constant. This motivates the definitions

$$\Theta(\zeta) := \left(\frac{\zeta z_\zeta(\zeta)}{\zeta^{-1} \bar{z}_\zeta(\zeta^{-1})} \right)^{1/2}, \quad (5.0.31)$$

$$S(\zeta) := -\Theta'(\zeta) + \gamma z_\zeta(\zeta). \quad (5.0.32)$$

Analytic continuation of (5.0.30) to $|\zeta| > 1$ yields a Riccati-type equation

$$-\Theta'(\zeta) + q_1(\zeta) \Theta^2(\zeta) = q_2(\zeta), \quad (5.0.33)$$

with

$$q_1(\zeta) = \frac{\gamma \bar{z}_\zeta(\zeta^{-1})}{\zeta^2}, \quad q_2(\zeta) = \frac{W_\zeta(\zeta) W_\zeta(\zeta^{-1})}{2 \bar{z}_\zeta(\zeta^{-1})}. \quad (5.0.34)$$

Both coefficients $q_1(\zeta)$ and $q_2(\zeta)$ are analytic in the exterior of the unit circle. Equation (5.0.33) is therefore a Riccati equation with analytic coefficients. According to the standard theory of nonlinear differential equations (Hille [4]), solutions of Riccati equations can possess only movable singularities which are simple poles. Consequently, any movable singularity of $\Theta(\zeta)$ in $|\zeta| > 1$ must be a simple pole. Since $\Theta(\zeta)$ is defined in terms of the square root of $z_\zeta(\zeta)$, it follows that $z_\zeta(\zeta)$ can only have second-order poles in the exterior domain $|\zeta| > 1$.

This global singularity restriction strongly limits the analytic structure of the conformal mapping derivative. $z_\zeta(\zeta)$ is strongly constrained and ultimately leads to the rational form of $z_\zeta(\zeta)$.

For capillary waves, the coefficient $q_2(\zeta)$ has a simple decay behaviour as $\zeta \rightarrow \infty$. From (5.0.4) we have, $W(\zeta) = \frac{2\pi c}{k} + \frac{ic}{k} \log \zeta$, and hence,

$$W_\zeta(\zeta) = \frac{ic}{k\zeta}. \quad (5.0.35)$$

Substituting this into (5.0.34) and using (5.0.1), we see that $q_2(\zeta)$ contains an explicit factor ζ^{-1} . Moreover, as $\zeta \rightarrow \infty$ we have $\zeta^{-1} \rightarrow 0$, and since z_ζ is analytic near $\zeta = 0$, the reflected quantity $\bar{z}_\zeta(\zeta^{-1})$ remains bounded and tends to a finite non-zero limit. Therefore,

$$q_2(\zeta) \sim \frac{c_1}{\zeta}, \quad \zeta \rightarrow \infty. \quad (5.0.36)$$

for a constant c_1 . On the other hand, the Bernoulli condition on $|\zeta| = 1$, after analytic continuation to the exterior domain $|\zeta| > 1$, is equivalent to the Riccati-type equation

$$-\Theta'(\zeta) + q_1(\zeta)\Theta^2(\zeta) = q_2(\zeta). \quad (5.0.37)$$

Thus the left-hand side defines a function analytic in the exterior of the unit circle, and its asymptotic behaviour at infinity is determined by that of q_2 . Equivalently, introducing

$$S(\zeta) := -\Theta'(\zeta) + q_1(\zeta)\Theta^2(\zeta), \quad (5.0.38)$$

we obtain

$$S(\zeta) \text{ analytic in } |\zeta| \geq 1, \quad S(\zeta) \sim \frac{c_1}{\zeta}, \quad \zeta \rightarrow \infty. \quad (5.0.39)$$

Theorem 5.0.1. [9] *The Bernoulli free-boundary condition on $|\zeta| = 1$ is equivalent to the requirement that the function*

$$S(\zeta) := -\Theta'(\zeta) + q_1(\zeta)\Theta^2(\zeta) \quad (5.0.40)$$

is analytic in the exterior domain $|\zeta| \geq 1$ and satisfies the asymptotic condition

$$S(\zeta) \sim \frac{c_1}{\zeta}, \quad \text{as } \zeta \rightarrow \infty. \quad (5.0.41)$$

This provides a global analyticity reformulation of the Bernoulli free-boundary condition, replacing the boundary condition on $|\zeta| = 1$ by an analytic constraint in the exterior domain.

Seeking rational solutions consistent with the above structure leads to the form,

$$z_\zeta(\zeta) = \frac{iA}{\zeta} \left(\frac{\prod_{j=1}^N (\zeta - \eta_j)}{\prod_{k=1}^N (\zeta - \zeta_k)} \right)^2, \quad (5.0.42)$$

with $|\eta_j|, |\zeta_k| > 1$.

We can derive this general form of $z_\zeta(\zeta)$ by following the approach of Crowdy [9], based on the analytic structure of the conformal map and the associated auxiliary function.

We consider conformal maps of the form

$$z(\zeta) = \frac{2\pi}{k} + \frac{i}{k} (\log \zeta + f(\zeta)), \quad (5.0.43)$$

where f is analytic in $|\zeta| \leq 1$. Differentiating, we obtain

$$z_\zeta(\zeta) = \frac{i}{k} \left(\frac{1}{\zeta} + f'(\zeta) \right). \quad (5.0.44)$$

In particular, z_ζ always has a simple pole at $\zeta = 0$.

Introducing the function

$$\Theta(\zeta) = (\zeta z_\zeta(\zeta))^{1/2}. \quad (5.0.45)$$

From the Riccati formulation, $\Theta(\zeta)$ is analytic outside the unit circle and may have only *simple poles*. Consequently, $z_\zeta(\zeta)$ may only have *second-order poles* outside the unit circle.

Assume that z_ζ has a single such pole at $\zeta = \zeta_1$ with $|\zeta_1| > 1$. Then z_ζ must take the form

$$z_\zeta(\zeta) = \frac{g(\zeta)}{\zeta(\zeta - \zeta_1)^2}, \quad (5.0.46)$$

for some function $g(\zeta)$.

Since z_ζ is assumed to be rational and analytic at infinity, the only admissible singularity at infinity is a pole. The denominator has degree 3, hence $g(\zeta)$ must be a polynomial of degree at most 3.

Next, we determine the location and multiplicity of zeros of $g(\zeta)$. Since the conformal map must be locally one-to-one in $|\zeta| < 1$, we require that

$$z_\zeta(\zeta) \neq 0 \quad \text{for } |\zeta| < 1. \quad (5.0.47)$$

Hence all zeros of $g(\zeta)$ must lie outside the unit circle.

Moreover, since $\Theta(\zeta) = (\zeta z_\zeta(\zeta))^{1/2}$ must not develop branch points, zeros of z_ζ must occur with *even multiplicity*. Therefore $g(\zeta)$ must be a perfect square.

Combining these, the admissible form of z_ζ is

$$z_\zeta(\zeta) = \frac{A_1(\zeta - \eta_1)^2}{\zeta(\zeta - \zeta_1)^2}, \quad (5.0.48)$$

where $|\eta_1| > 1$.

We now use the condition that the auxiliary function $S(\zeta)$ is analytic in $|\zeta| \geq 1$. This would require the apparent second-order pole at $\zeta = \zeta_1$ to be removable. Therefore, both the coefficients of $(\zeta - \zeta_1)^{-2}$ and $(\zeta - \zeta_1)^{-1}$ must vanish.

Carrying out this calculation gives the relations

$$\eta_1 = -\zeta_1, \quad (5.0.49)$$

together with a second condition relating parameters.

Next, we use the asymptotic condition at infinity,

$$S(\zeta) \sim \frac{c_1}{\zeta}, \quad \text{as } \zeta \rightarrow \infty. \quad (5.0.50)$$

This determines the remaining parameter and gives

$$A_1 = iA, \quad (5.0.51)$$

for some real constant A .

Thus we get

$$z_\zeta(\zeta) = \frac{iA(\zeta - \eta_1)^2}{\zeta(\zeta - \zeta_1)^2}. \quad (5.0.52)$$

Integrating,

$$z(\zeta) = \frac{2\pi}{k} + iA \left(\log \zeta - \frac{4\zeta_1}{\zeta - \zeta_1} \right), \quad (5.0.53)$$

The general case with N poles follows by the same reasoning: allowing N second-order poles ζ_k , enforcing even multiplicity of zeros, and preserving the factor $1/\zeta$, we get

$$z_\zeta(\zeta) = \frac{iA}{\zeta} \left(\frac{\prod_{j=1}^N (\zeta - \eta_j)}{\prod_{k=1}^N (\zeta - \zeta_k)} \right)^2, \quad |\eta_j|, |\zeta_k| > 1. \quad (5.0.54)$$

Imposing removability of poles of $S(\zeta)$ fixes the parameters algebraically. For $N = 1$, one obtains

$$z(\zeta) = \frac{2\pi}{k} + \frac{i}{k} \left(\log \zeta - \frac{4\zeta_1}{\zeta - \zeta_1} \right), \quad (5.0.55)$$

which matches exactly with Crapper's classical solution. Higher N correspond to multi-periodic generalizations.

The Riccati equation implies that $\zeta = \infty$ is a cluster point of poles. Consequently, solutions admit a pure Weierstrass product representation

$$z_\zeta = \frac{A}{\zeta^2} \left(\frac{\prod_j (\zeta - \eta_j)}{\prod_j (\zeta - \zeta_j)} \right)^2. \quad (5.0.56)$$

A refined counting argument shows that two additional zeros must occur outside this pairing.

The central conceptual message of the reformulation in Crowdy's paper [9] is that the free-surface Bernoulli condition is equivalent to a statement about global analyticity. This viewpoint is the foundation for later developments in the analytic continuation of free boundaries into unphysical domains, and directly informs modern singularity-based approaches to water-wave theory.

Chapter 6

Further Research directions

6.1 Singularities of Water Waves with Vorticity

The analysis in this thesis concerns steady periodic gravity waves in the case of irrotational flow. Under this assumption the velocity field u satisfies

$$\nabla \times u = 0, \quad \nabla \cdot u = 0, \quad (6.1.1)$$

and therefore can be written in terms of a velocity potential

$$u = \nabla \phi, \quad (6.1.2)$$

where ϕ satisfies Laplace's equation

$$\Delta \phi = 0 \quad (6.1.3)$$

in the fluid domain.

This harmonic structure allows the use of complex analytic methods and conformal mappings to study the analytic continuation of the free surface and the associated singularities in the complex plane. In particular, the analysis carried out in Chapters 4 and 5 shows that the analytically continued solution develops singularities of square-root type in the complex plane, corresponding to branch point singularities of the conformal mapping.

An important extension of this theory is the study of steady water waves in the presence of vorticity. In this case the velocity field satisfies

$$\nabla \times u = \omega, \quad (6.1.4)$$

where ω denotes the vorticity of the flow. The incompressibility condition still implies

$$\nabla \cdot u = 0. \quad (6.1.5)$$

In two dimensions it is convenient to introduce a stream function ψ such that

$$u = (\psi_y, -\psi_x). \quad (6.1.6)$$

The vorticity is then given by

$$\omega = -\Delta \psi. \quad (6.1.7)$$

For flows with prescribed vorticity distribution $\omega = \gamma(\psi)$, the governing equation becomes the nonlinear elliptic equation

$$\Delta\psi + \gamma(\psi) = 0. \quad (6.1.8)$$

This replaces the Laplace equation that arises in the irrotational case. The free surface boundary conditions take the form

$$\psi = \text{constant} \quad \text{on the free surface,} \quad (6.1.9)$$

together with the Bernoulli condition

$$\frac{1}{2}|u|^2 + gy = B, \quad (6.1.10)$$

where B is a constant.

The presence of vorticity fundamentally changes the analytic structure of the problem. In the irrotational setting in this thesis, the harmonicity of the velocity potential allows the use of complex analytic continuation and leads to singularities of square-root type in the complex plane. When vorticity is present, the governing equation becomes nonlinear and the harmonic structure is lost, making the analytic continuation problem a bit more complicated.

In particular, the singularity structure obtained in Chapters 4 and 5 arises from differential equations governing the analytically continued variables, whose local solutions have expansions of the form

$$F_1(\zeta) = \frac{1}{(\zeta - \zeta_0)^{1/2}} \sum_{n=0}^{\infty} \tilde{b}_n (\zeta - \zeta_0)^{n/2}. \quad (6.1.11)$$

This expansion shows that the analytically continued solution possesses branch point singularities of square-root type. When vorticity is present, the nonlinear equation

$$\Delta\psi + \gamma(\psi) = 0 \quad (6.1.12)$$

replaces the harmonic equation and may significantly change the analytic continuation of the solution.

Nevertheless, conformal mapping techniques can still be applied in certain cases, particularly when the vorticity is constant. In such cases, the governing equations reduce to nonlinear relations for the conformal mapping and the complex velocity that depend explicitly on the vorticity parameter (see, for example, [13, 14, 16]). The analytic continuation of these equations may then lead to singularities whose structure depends on the vorticity distribution, and related phenomena have also been studied in more general vorticity settings (see [15]).

An important open problem would be to determine how the presence of vorticity influences the location and type of singularities in the complex plane. In particular, we can ask whether the branch-point singularities in the irrotational case also exist when $\gamma(\psi) \neq 0$, or whether additional types of singularities are formed due to the nonlinear vorticity term.

Understanding the interaction between vorticity and complex singularities therefore is an important direction for further research. Extending the singularity analysis developed in this

thesis to rotational flows could provide new insight into the analytic structure of steady water waves and the role of vorticity in determining the behaviour of the free surface.

Bibliography

- [1] J. K. Hunter, *An Introduction to the Incompressible Euler Equations*, Lecture Notes, University of California, Davis, September 2006.
- [2] T. W. Gamelin, *Complex Analysis*, Undergraduate Texts in Mathematics, Springer-Verlag, New York, 2001.
- [3] H. Okamoto and M. Shōji, *The Mathematical Theory of Permanent Progressive Water-Waves*, World Scientific, Singapore, 2001.
- [4] E. Hille, *Ordinary Differential Equations in the Complex Domain*, Wiley, New York, 1976, Chapter 3.
- [5] R. S. Johnson, *A Modern Introduction to the Mathematical Theory of Water Waves*, Cambridge University Press, Cambridge, 1997.
- [6] E. L. Ince, *Ordinary Differential Equations*, Dover Publications, New York, 1956.
- [7] H. Lewy, *A Note on Harmonic Functions and a Hydrodynamical Application*, Proceedings of the American Mathematical Society, 3 (1952), 111–113.
- [8] S. Tanveer, *Singularities in water waves and Rayleigh–Taylor instability*, Proceedings of the Royal Society A, 435 (1991), 137–158.
- [9] D. G. Crowdy, *A new approach to free surface Euler flows with capillarity*, Journal of Fluid Mechanics, 953 (2000), A35.
- [10] A.-M. Matei, *The Neumann problem for free boundaries in two dimensions*, C. R. Acad. Sci. Paris, Ser. I 335 (2002), 1–6.
- [11] G. D. Crapper, *An exact solution for progressive capillary waves of arbitrary amplitude*, Journal of Fluid Mechanics, 2 (1957), 532–540.
- [12] G. R. Baker and C. Xie, *Singularities in the complex physical plane for deep water waves*, Journal of Fluid Mechanics, 685 (2011), 83–116.
- [13] D. G. Crowdy and J. Roenby, *Analytical solutions for periodic deep-water waves with constant vorticity*, European Journal of Mechanics B/Fluids, 46 (2014), 179–188.
- [14] Hur, V.M. & Wheeler, M.H. *2020 Exact free surfaces in constant vorticity flows. J. Fluid Mech.* 896, R1.
- [15] J. Weber and E. Wahlén, *Global bifurcation of capillary-gravity water waves with overhanging profiles and arbitrary vorticity*, Communications in Mathematical Physics, 382 (2021), 123–153.
- [16] F. Carvalho, *Gravity water waves over constant vorticity flows: from laminar flows to touching waves*, Master’s thesis, Centre for Mathematical Sciences, Lund University, 2023.